

Proposal template Part B: technical description



RIPPLE: REDUCING IMPACT OF PLASTIC POLLUTION THROUGH LEARNING & ENGAGEMENT

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1.Excellence #@REL-EVA-RE@#

Plastic pollution is one of the fastest-growing threats to marine and freshwater ecosystems, with more than 80% of marine litter originating from land-based sources (UNEP, 2016; EEA, 2022). Rivers are the main pathways, carrying plastic from cities, industries, and agriculture into seas and coastal zones, yet monitoring and prevention remain fragmented and inconsistent across Europe. The new guidelines of the MSFD Technical Group on Marine Litter (TG ML, 2023) explicitly call for harmonised monitoring methods across all compartments (beach, surface, seafloor, micro/macro), and for open, standardised data flows into European infrastructures such as EMODnet and the Digital Twin of the Ocean (DTO). RIPPLE advances beyond this fragmented landscape by developing and demonstrating a **Mediterranean Toolbox** that integrates harmonised monitoring protocols, validated removal technologies, societal prevention measures, and circular innovation. Unlike existing initiatives that focus on isolated pilots or awareness campaigns, RIPPLE combines these elements into a coherent, authority-owned

framework that is interoperable with EU data infrastructures and ready for replication across Mediterranean basins. On monitoring, the project introduces low-cost in-situ sensors, drones with computer vision, and citizen science protocols aligned with TG ML, with datasets validated and published within three months through EMODnet/DTO. This directly addresses the urgent gaps in inland and riverine plastic monitoring identified by the EEA (2022). The project will map existing innovative technologies for waste removal and manipulation and create an evaluation and selection framework for municipalities. Furthermore through community driven co-design sessions with citizens and local creatives the project will pilot and validate selected solutions and design use cases and implementation strategies for the municipalities. Each will undergo life-cycle assessment and ecotoxicity testing, ensuring environmental safety and cost-effectiveness, filling a major gap in today's evidence base where devices are often deployed without robust validation (MSFD TG ML, 2023). On prevention, the project builds on proven approaches highlighted in OSPAR and Mission Ocean projects such as INSPIRE and REMEDIES, but goes further by embedding them into citizen behaviour-change programmes. The integration of social innovation and design-thinking through makerspaces and FabLabs adds a novel dimension: plastics collected are not only removed but transformed into new products through open source circular technologies, such as Precious Plastics, mobile FabLabs, digital fabrication machines, linking environmental protection with local entrepreneurship and education. Finally, RIPPLE ensures basin-level impact by replicating solutions through an **Associated Regions (ARs) scheme**, offering cascade funding, structured mentoring, and technical assistance. This mechanism — not yet present in marine litter projects — guarantees that solutions demonstrated in Spain, Greece, and Croatia will be adapted and transferred to additional Mediterranean authorities, creating system-wide impact. The project's ambition lies in delivering an integrated, TRL 7–8, authority-ready Toolbox of real-world solutions spanning land-river-sea. The Toolbox will integrate MSFD TG ML-compliant monitoring protocols, supplemented by fast open data streams into EMODnet and the Digital Twin of the Ocean; tested removal technologies validated under realistic riverine and coastal circumstances; systemic prevention measures co-designed with municipalities & citizens as well as circular reuse pathways turning collected plastics back to resources.. By combining technical innovation with social engagement and open science, RIPPLE makes a step-change in how Mediterranean regions can achieve the Mission Ocean target of a **50% reduction in plastic pollution by 2030**.

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

The project aims to co-design, demonstrate, and scale a Toolbox of solutions that enables public authorities to systematically prevent, monitor, and remove plastic litter along the land–river–sea continuum. The Toolbox will integrate four essential components: (i) harmonised monitoring methods fully aligned with the MSFD Technical Group on Marine Litter (TG ML) guidelines, delivering open and quality-controlled datasets to EMODnet and the EU Digital Twin Ocean; (ii) validated low-cost open source removal technologies, tested under real riverine and coastal conditions to ensure environmental safety and cost-effectiveness; (iii) prevention and uptake measures, co-developed with municipalities, stakeholders, local creative communities (makers and technology innovators), and citizens to address systemic pollution; and (iv) educational and circular innovation programmes for municipalities to empower schools, communities, and social entrepreneurs in transforming collected plastics into resources. The objectives are directly pertinent to the Work Programme topic HORIZON-MISS-2025-03-OCEAN-02, which calls for basin-level solutions embedding monitoring, removal, and societal engagement. The project will explicitly address the Mediterranean basin through three demonstration sites in Spain, Greece, and Croatia, where Toolbox modules will be implemented and tested. Monitoring at these sites will provide measurable, verifiable evidence of reduced plastic fluxes (items/km, kg/day, and percentage reduction of leakage into the sea). The ambition of the project lies in validating, integrating, and operationalising existing proven solutions into a single authority-owned Toolbox that is interoperable with existing European infrastructures (EMODnet, DTO) and accompanied by open playbooks, evaluation frameworks, procurement templates, and replication strategies. This ensures long-term usability for municipalities, regional agencies, and water managers. The project is ambitious yet realistic for a 48-month Innovation Action. It builds on solutions and prototypes currently at TRL 5–6 (e.g. Floating Booms, Trash Traps, Bio barriers, Precious Plastics, mobile FabLabs, citizen science protocols, digital monitoring tools) and will advance them to TRL 7–8 by project end, delivering operational, user-ready use-cases. Beyond the three pilots, the project will engage at least three ARs through cascade funding, mentoring, and technical assistance, ensuring basin-level replication and capacity building. By consolidating technical, social, digital, and educational innovations into one coherent Toolbox, the project contributes directly to the Mission Ocean and Waters targets, including the 50% reduction of plastics in the sea by 2030, the EU Zero Pollution Action Plan, and the development of the Digital Ocean Knowledge System.

1.1.1 Specific objectives

At its core, the project seeks to integrate scientific monitoring, technical innovation, and social engagement to address the persistent challenge of plastic pollution along the land–river–sea continuum. The project ensures that interventions are not only environmentally sound, but also socially inclusive. At its core, the RIPPLE seeks to integrate scientific monitoring, technical innovation, and social engagement to tackle plastic pollution along the land–river–sea continuum in the Mediterranean basin. By embedding monitoring, removal, prevention, and education into a single Toolbox, it ensures that interventions are environmentally sound, socially inclusive, and directly usable by public authorities. In line with Horizon Europe requirements, the project will actively cooperate with the Mission Ocean & Waters Implementation Platform, the Plastic Pirates campaign, the WATER4ALL Partnership, and relevant Interreg initiatives, while also aligning with the Zero Pollution Action Plan, the New European Bauhaus, and the European Green Deal. The project defines seven Specific Objectives (SOs), each aligned with the expected outcomes of the call and supported by Technical Objectives (TOs) that link directly to WPs and tasks.

S.O.1 – Co-design prevention and collection measures with authorities, industry, and citizens This objective ensures prevention is embedded at source, with ownership by municipalities, citizens, and relevant industries.

T.O.1.1 Engage local ecosystems to identify risks and co-design integrated solutions (T1.1-T1.3; T2.1-T2.2).

T.O.1.2 Run citizen and school programmes, awareness campaigns, and behavioural nudges (T1.3; T2.4; T3.3).

T.O.1.3 Co-develop procurement templates and waste-system improvements with municipalities (T2.5; T4.2).

S.O.2 – Empower municipalities and communities with education and circular solutions This objective ensures behavioural change is sustained through training and circular entrepreneurship.

T.O.2.1 Develop sustainability education programmes for municipalities, schools, and citizens (T3.1–T3.3).

T.O.2.2 Integrate citizen science and co-creation methodologies (T1.2; T2.2).

T.O.2.3 Enable social entrepreneurship for circular transformation of collected plastics (T2.4; T3.3).

S.O.3 – Establish harmonised monitoring and data systems for river-to-sea plastics This objective secures robust evidence and interoperability with EU infrastructures.

T.O.3.1 Develop MSFD TG ML-compliant monitoring protocols (T5.3; 5.4).

T.O.3.2 – Implement data pipelines to EMODnet and validate DTO interoperability (T5.1–T5.2).

T.O.3.3 – Deliver open datasets within three months, incl. metadata and QA/QC (T5.1; T5.2; T5.3; T5.4).

S.O.4 – Demonstrate sustainable and safe removal technologies The project will validate innovative removal devices under real-world conditions, ensuring no harm to ecosystems.

T.O.4.1 Pilot and validate removal technologies in riverine hotspots (T2.3, T6.1 - T6.4).

T.O.4.2 Validate environmental safety and efficiency of removal solutions via LCA and ecotoxicity (T5.3; T5.4).

T.O.4.3 Produce operational playbooks and O&M guidelines for public authorities (T4.1–T4.4).

S.O.5 – Demonstrate the Toolbox in three Mediterranean use-cases Spain, Greece, and Croatia will serve as demonstration pilots to test integrated solutions.

T.O.5.1 Implement Toolbox modules in the three use-case sites (T6.1 - T6.4).

T.O.5.2 Conduct baseline and endline monitoring to deliver quantified reductions (T5.3–T5.4; T6.2).

T.O.5.3 Publish site-level impact reports and datasets (T5.4; T8.3).

S.O.6 – Replicate solutions through ARs Replication ensures basin-level impact beyond pilots.

T.O.6.1 Design and implement the ARs scheme (T7.4).

T.O.6.2 Provide mentoring, technical assistance, and capacity building (T1.3; T7.4).

T.O.6.3 Deliver replication roadmaps and uptake strategies (T4.3; T7.4; T7.5).

S.O.7 – Ensure policy impact, exploitation, and long-term sustainability This objective secures adoption of results and pathways for exploitation beyond the project.

T.O.7.1 Produce adoption roadmaps, policy briefs, and standardisation inputs (T7.3).

T.O.7.2 Develop financing and procurement pathways (T7.2; T7.3; T7.5).

T.O.7.3 Deliver an exploitation strategy to secure sustainability (T7.2; T7.3, T7.5).

Each Specific Objective is directly linked to the Expected Outcomes of the call HORIZON-MISS-2025-03-OCEAN-02, ensuring that monitoring and open data (SO3), removal solutions (SO4), prevention and societal engagement (SO1, SO2), demonstration in three Mediterranean use-cases (SO5), replication via ARs (SO6), and policy/exploitation (SO7) are all fully addressed.

SMART Self-Assessment Canvas for CURBSCAPES Technical Objectives

No.	Specific	Measurable	Attainable	Relevant	Time-Bound
T.O.1.1	Engage local ecosystem to identify risks and	# of stakeholder workshops; # of	Feasible with municipalities,	Anchors prevention in governance &	First workshops by

	co-design integrated prevention & collection solutions	integrated solutions co-designed	NGOs, SMEs	community practices	M6; solutions designed by M18
T.O.1.2	Run citizen and school programmes, awareness campaigns, behavioural nudges	# of campaigns; # of schools engaged; citizen participation levels	Partners already run educational & awareness programmes	Builds long-term behavioural change & prevention	Campaigns launched by M18; ongoing annually
T.O.1.3	Co-develop procurement templates & waste-system improvements with municipalities	# of templates delivered; # of municipalities adopting improvements	Feasible with policy/municipality partners	Ensures systemic prevention & uptake	Templates ready by M24; tested by M30
T.O.2.1	Develop sustainability education programmes for municipalities, schools & citizens	# of training modules; participant numbers	Universities and education partners already experienced	Builds local authority & citizen capacity	Framework by M12; modules piloted by M36
T.O.2.2	Integrate citizen science & co-creation methodologies	# of citizen protocols developed; # datasets generated	Supported by science & civic partners	Increases data coverage & legitimacy	Protocols by M12; applied in pilots by M33
T.O.2.3	Enable social entrepreneurship for circular transformation of plastics	# of pilots; # of entrepreneurs engaged	FabLabs and SMEs in consortium already active	Creates circular economy dimension of Toolbox	First pilots by M33; results by M43
T.O.3.1	Develop MSFD TG ML-compliant monitoring protocols	% compliance with TG ML; # stations deployed	RTOs & marine science partners capable	Adapting and operationalising existing TG ML protocols, Provides scientific backbone of Toolbox	Protocols by M12; full deployment by M18
T.O.3.2	Implement data pipelines to EMODnet & validate DTO interoperability	# datasets published; DTO integration tests passed	Data/ICT partners (ICCS, TechnAi) have expertise	Ensures open science & EU legacy	First data by M12; Full data flows following M18; DTO validation by M24
T.O.3.3	Deliver open datasets within 3 months of acquisition incl. QA/QC	Time to publication ≤ 3 months; # datasets	Consortium has QA/QC protocols	Guarantees transparency & comparability	Continuous; first release by M12
T.O.4.1	Pilot & validate removal technologies (booms, traps, robotics, bins)	% flux reduction; kg/day removed	SMEs & pilots already identified	Demonstrates technical viability of Toolbox	Pilots by M33; evaluation by M36
T.O.4.2	Validate environmental safety & efficiency (LCA, ecotoxicity, €/kg removed)	# LCAs; ecotoxicity reports	Consortium has LCA expertise (CSIC etc.)	Ensures no-harm & cost-effectiveness	Assessments by M45
T.O.4.3	Produce O&M playbooks & guidelines for authorities	# playbooks delivered; feedback from pilots	Feasible via public authority partners	Critical for adoption beyond project	Playbooks by M30
T.O.5.1	Implement Toolbox modules in 3 use-case	# of modules deployed; # of	Pilots secured in Spain,	Validates Toolbox in real conditions	Deployment start by M18;

	sites	authorities engaged	Greece, Croatia		results by M33
T.O.5.2	Conduct baseline & endline monitoring with reduction results	% reduction; kg/day, items/km	MSFD-compliant methods embedded	Provides measurable outcomes for Mission target	Baseline by M18; endline by M43
T.O.5.3	Publish site-level impact reports & datasets	# reports published; # datasets open	Supported by monitoring/data partners	Ensures transparent communication of impact	Reports by M48
T.O.6.1	Design & implement ARs scheme	# of regions selected; open call launched	FSTP rules + partner governance structure in place	Key element of the call for replication	Scheme launched by M24; regions onboard by M33
T.O.6.2	Provide mentoring, technical assistance, capacity building	# training sessions; # mentoring hours	Feasible with universities & SMEs	Ensures replication readiness	Ongoing from M18; at least 3 regions by M33
T.O.6.3	Deliver replication roadmaps & uptake strategies	# roadmaps delivered; # regions with uptake plans	Partners experienced in replication frameworks	Extends impact beyond pilots	Roadmaps by M33
T.O.7.1	Produce adoption roadmaps, policy briefs, standardisation inputs	# briefs produced; # stakeholders reached	Universities & networks capable	Ensures EU/national policy impact	First briefs by M24; full set by M43
T.O.7.2	Develop financing & procurement pathways	# pathways published; # PA adopting	Feasible via exploitation partners	Supports sustainability beyond funding	Pathways by M30
T.O.7.3	Deliver exploitation strategy for long-term sustainability	Exploitation plan published; commitments secured	Consortium already active in exploitation	Critical for post-project continuation	Strategy by M33

The project will deliver a coherent set of results that transform fragmented approaches into an integrated Mediterranean Toolbox, directly contributing to the call's Expected Outcomes. The results are specific, measurable, and anchored in the pilot demonstrations, while also designed for basin-wide replication.

R1 – Mediterranean Toolbox: An operational, authority-owned Toolbox integrating monitoring protocols, validated removal technologies, prevention measures, education modules, and replication frameworks, aligned with MSFD TG ML guidelines and interoperable with EMODnet and the EU Digital Twin Ocean. **R2 – Harmonised monitoring datasets:** Open, quality-controlled datasets on land–river–sea plastic flows (kg/day, items/km, % flux reduction), published within three months of acquisition. Full compliance with FAIR principles, validated for DTO interoperability. **R3 – Validated removal technologies:** Demonstrated and environmentally safe solutions (such as Floating Booms, Trash Traps, Bio barriers), tested under river-to-sea conditions with full LCA and ecotoxicity assessments, ensuring cost-effectiveness and no-harm safeguards. **R4 – Operational playbooks and procurement templates:** Practical guidance for municipalities and public authorities on deployment, operation, and maintenance of removal and prevention solutions, including standardised procurement pathways. **R5 – Education and training programmes:** Tailored curricula and toolkits for municipalities, schools, and communities, integrating citizen science and sustainability education, tested in pilots and ready for wider replication. **R6 – Circular entrepreneurship pilots:** Local FabLabs and makerspaces demonstrating transformation of collected plastics into usable products, generating business cases for social enterprises and community-led ventures. **R7 – Demonstrated flux reduction at pilot sites:** Measurable reductions in plastic litter leakage at three Mediterranean use-cases (Spain, Greece, Croatia), with baseline and endline monitoring providing quantified impact data. **R8 – Replication roadmaps for ARs:** Structured mentoring, capacity building, and technical assistance for ≥ 3 ARs, supported by cascade funding, enabling replication of Toolbox solutions beyond pilots. **R9 – Policy, adoption, and exploitation outputs:** Roadmaps, policy briefs, and standardisation contributions linked to the Zero Pollution Action Plan, Mission

Ocean, and Green Deal strategies, ensuring long-term sustainability and basin-level policy impact.

Result	Linked SO/TO	Relevant Tasks	Expected Outcome (EO) Link
R1	SO4, SO5, SO7	T4.1–T4.5; T6.1–T6.4; T5.3; T5.4; T8.3; T7.3 - T7.5	EO1 Toolbox for authorities
R2	SO3 (TO3.1–3.3)	T5.1–T5.4	EO4 Enhanced knowledge, EO6 DTO/EMODnet
R3	SO4 (TO4.1–4.2)	T6.1–T6.4; T5.3; T5.4	EO2 Reduced plastic leakage, EO3 Innovative removal
R4	SO1(TO1.3), SO4(TO4.3), SO7(TO7.2)	T2.5; T4.2–T4.4; T7.2; T7.3; T7.5	EO3 Uptake of solutions
R5	SO2 (TO2.1–2.2), SO6	T3.1–T3.3; T1.2; T2.2; T4.3; T1.3; T7.4; T7.5	EO3 Societal engagement
R6	SO2 (TO2.3), SO1 (TO1.2)	T2.4; T3.3; T2.4; T1.3	EO3 Prevention & uptake
R7	SO4 (TO4.1–4.3), SO5 (TO5.2)	T6.1–T6.4; T5.3; T5.4;	EO2 Reduction in rivers & seas
R8	SO5 (TO5.1–5.3), SO7	T6.1–T6.4; T5.3; T5.4; T8.3; T7.2–T7.5	EO3 Replication & scaling
R9	SO7 (TO7.1–7.3)	T7.2–T7.3; T7.5	EO5 Mission Ocean & Zero Pollution targets

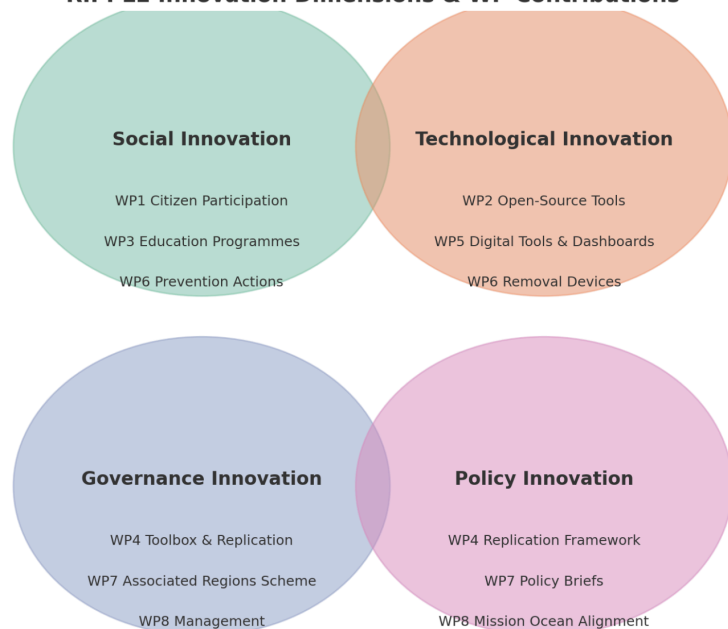
1.1.2 Beyond the state-of-the-art

The project goes beyond the state of the art by delivering, for the first time, an integrated Mediterranean Toolbox that combines harmonised monitoring, validated removal, systemic prevention, education, and structured replication in a single authority-owned package. Existing efforts remain fragmented: inland monitoring is sporadic and inconsistent; removal devices are tested without ecological validation; and prevention measures are disconnected from the citizens and the local communities. Our Toolbox fills these gaps by embedding MSFD TG ML (2023)–compliant protocols and ensuring all datasets are quality-controlled, openly available, and routed through EMODnet with full interoperability to the EU Digital Twin Ocean (DTO). This ensures basin-wide comparability and further standardised data flows to correspond with EU requirements. Despite regional datasets collected by singular pilots, there is no integrated initiative that has harmonised monitoring to this level and serves as an operational system with digital interoperability of sea status information across the Mediterranean basin. Through addressing this deficiency, the initiative provides a scientific and technological platform for driving unrelenting plastic pollution reduction in support of incipient cross-border policy action. On removal technologies, current innovation is advancing rapidly, but few devices have been validated under river-to-sea conditions or undergone systematic life-cycle and ecotoxicity assessments. Our pilots will couple trials of low-tech low-cost open source solutions, such as Floating Booms, Trash Traps, Bio barriers, with LCA and safety testing, producing validated solutions and operational playbooks that municipalities can adopt without repeating fragmented experiments. On monitoring, reviews consistently highlight constraints in coverage, QA/QC, and comparability. We advance this state of the art by deploying multi-source approaches — low-cost in-situ sensors, drone/computer-vision transects, and structured citizen science protocols. All methods will be mapped to MSFD TG ML guidelines and deliver quality-controlled datasets within three months of acquisition. This guarantees not only scientific credibility but also policy usability, as the data stream will directly support Good Environmental Status tracking and reporting under the Marine Strategy Framework Directive. On prevention and uptake, regional initiatives (e.g. OSPAR, INSPIRE, REMEDIES) confirm that various stakeholders in the ecosystem (such as ports, fisheries, and schools) can play a role, but efforts are uneven and rarely embedded in authority-led toolkits. This replication mechanism, supported by cascade funding (€100k per region), mentoring, and technical assistance, is novel in marine litter projects and ensures that results travel beyond the three pilots.

Innovation dimension RIPPLE introduces a systemic innovation model that goes beyond incremental advances in marine litter management. While existing projects address monitoring, removal, or awareness in isolation, RIPPLE integrates social, technological, governance, and policy innovations into a single Mediterranean Toolbox. Social innovation is embedded through citizen science protocols and community co-design; technological innovation emerges from combining low-cost circular tools with advanced digital dashboards interoperable with the DTO; governance innovation is piloted through the Associated Regions scheme and procurement templates for municipalities; and policy innovation ensures direct contribution to Mission Ocean targets. The novelty lies not only in each component, but in their integration: an end-to-end pathway from prevention to replication that disrupts fragmented approaches and enables systemic, scalable impact. In summary, the ambition of the project is not to create another isolated gadget or campaign, but to deliver a basin-ready, authority-owned integration of harmonised

monitoring, validated removal, systemic prevention, and open data interoperability with EMODnet/DTO. By advancing key solutions from TRL 5–6 to TRL 7–8 and simultaneously raising their societal readiness through education and citizen engagement, the project provides a step-change in how Mediterranean regions can realistically achieve the Mission Ocean target of a 50% reduction in plastic pollution.

RIPPLE Innovation Dimensions & WP Contributions



1.1.3 R&I Maturity and Technology Readiness Levels (TRL)

The project is positioned at the innovation and demonstration stage, advancing promising solutions for marine litter monitoring, removal, and prevention from mid-level maturity (TRL 4–6) to validated, pre-operational readiness (TRL 7–8) in real environments. Unlike exploratory research actions, this Innovation Action focuses on scaling and integrating solutions into a Mediterranean Toolbox designed for immediate use by public authorities and communities. At the start of the project, most technical solutions are already beyond the idea stage but remain fragmented and insufficiently validated: **Monitoring technologies** (low-cost in-situ sensors, drone/computer vision, citizen science protocols) are at TRL 4–5, with pilot trials but not yet harmonised or interoperable with EU systems; **Removal technologies** (such as

Floating Booms, Trash Traps, Bio barriers,) are at TRL 6–8, with prototypes tested locally but lacking full LCA/ecotoxicity validation under river-to-sea conditions; **Digital integration** tools (dashboards, data pipelines to EMODnet, DTO interoperability) currently stand at TRL 5, with partial demonstrations in pilot settings but not yet fully operational at basin scale. Through structured demonstration in three Mediterranean pilot sites (Spain, Greece, Croatia), these solutions will be validated, standardised, and integrated. By the end of the project: Monitoring protocols and datasets will reach TRL 7–8, feeding directly into EMODnet and the Digital Twin Ocean within three months of acquisition; Removal technologies will reach TRL 7–9, with validated safety, efficiency, and cost-effectiveness, supported by operational playbooks for municipalities; Digital dashboards and the Toolbox platform will reach TRL 7–8, enabling real-time decision-making and ensuring replication in ARs. In parallel, the project advances the Societal Readiness Level (SRL) by embedding education programmes, citizen science, and sector agreements (e.g., schools, citizen groups, social enterprises, fisheries, ports) to ensure uptake. At project start, prevention and educational practices sit at SRL 3–4 (tested in local initiatives); by project end they will reach SRL 6–7, with authority-owned curricula, procurement templates, and replication strategies embedded in governance processes. This dual advancement of TRL and SRL ensures that the project's ambition is realistic yet transformative: solutions move from fragmented pilots to integrated, authority-owned, and replication-ready tools that directly support the Mission Ocean target of reducing plastic pollution by 50%.

Monitoring and Data Systems Although marine litter monitoring has advanced at sea, inland rivers and deltas still lack harmonised, systematic approaches. The project will close this gap by deploying cost-effective, MSFD TG ML-compliant monitoring systems, ensuring comparability across sites, and by establishing direct data flows to EU infrastructures such as EMODnet and the Digital Twin of the Ocean, enabling basin-scale integration and policy uptake..

Existing solutions in pilots: Drone transects for surface litter (Spain), community sampling (Greece), in-situ sensors for river plastics (Croatia).

Possible solutions to be developed: AI-driven litter detection from drones (TRL 5→7), smart sensor kits for schools and citizen science (TRL 4→6), real-time data pipelines to EMODnet/DTO (TRL 5→8).

Removal Technologies Innovative removal devices are proliferating, but most lack ecological validation or cost-benefit analysis. The project will validate and optimize solutions to ensure no-harm and uptake readiness.

Existing solutions in pilots: Floating booms and traps (Spain), robotic skimmers (Greece), river barriers (Croatia).

Possible solutions to be developed: Floating Barriers (TRL 7–9), Trash Traps (TRL 6–8), Bio barriers TRL 5–7 , integrated playbooks with LCA/ecotoxicity validation.

Prevention and Uptake Prevention measures often stop at awareness campaigns without structural uptake. The project integrates societal, sectoral, and governance dimensions.

Existing solutions in pilots: Fishing-for-litter schemes (Greece), municipal waste fixes (Spain), citizen campaigns (Croatia).

Possible solutions to be developed: Procurement templates for waste operators (TRL 4→7), behavioural nudges with digital tools (TRL 5→7), port reception system upgrades (TRL 5→8).

Circular Transformation and Social Entrepreneurship Plastic waste streams are rarely valorised locally. The project activates circular pilots that empower citizens and SMEs.

Existing solutions in pilots: Precious Plastic machines (Croatia), maker spaces for upcycling (Spain), school-led plastic reuse (Greece).

XR-enabled education programs connect plastic pollution monitoring with circular economy action, moving beyond awareness to practical community-driven solutions to establish a transferable model that empowers municipalities, schools, and citizens to become active agents of environmental change.

Possible solutions to be developed: Mobile FabLabs for rural/river sites (TRL 4→7), open-source digital machinery for plastic-to-product (TRL 5→7), entrepreneurship models connecting waste valorisation with local markets.

Education and Capacity Building Awareness without structured education rarely produces systemic change. The project embeds circularity and prevention into formal and informal education.

Existing solutions in pilots: School programmes on marine litter (Spain), NGO awareness activities (Croatia).

Possible solutions to be developed: Mediterranean education framework for municipalities (TRL 4→6), digital training modules integrated into Toolbox (TRL 5→7), experiential “learning by doing” programmes with citizen science kits.

Replication and ARs Replication is often left to the final stage of projects, limiting impact. This project ensures early and continuous involvement of ARs.

Existing solutions in partners: Regional networks for waste management (Croatia, Greece).

Possible solutions to be developed: Mentoring and technical assistance packages (TRL 5→7), open call funding for ≥3 ARs (TRL 6→8), replication roadmaps embedded in Toolbox.

Solution Cluster	TRL/SRL at Start	TRL/SRL at End	Key Advancement Activities
Monitoring tools & methods	TRL 4–5 (lab prototypes, small pilots)	TRL 7-8 (field validated, integrated with EMODnet & DTO)	Develop MSFD TG ML–compliant protocols; deploy sensors/drones; integrate citizen science; publish open datasets within 3 months.
Removal technologies	TRL 5–8 (tested in controlled environments)	TRL 7–9 (validated in real river–delta–coastal sites)	Pilot low-tech, low-cost open source collection solutions; conduct LCA & ecotoxicity analyses; produce operational playbooks for public authorities.
Digital tools & platform	TRL 4–5 (prototype dashboards & pipelines)	TRL 7-8 (integrated, user-ready services with DTO/EMODnet)	Develop dashboards & forecasting tools; validate data pipelines; ensure interoperability; deliver Toolbox digital platform.
Education & circular innovation	SRL 3–4 (local practices, training pilots)	SRL 6–7 (validated curricula, authority-owned roadmaps)	Design and test education programmes; co-create citizen guides; pilot plastic reuse entrepreneurship; develop replication strategies.

#\$PRJ-OBJ-PO\$#

1.2 Methodology #@CON-MET-CM@# #@COM-PL-CP@#

1.2.1 Concept and methodology

Starting from the recognition that plastic pollution is not only an ecological issue but also a governance, societal, and economic challenge, the project will support Mediterranean river–sea territories in enhancing their capacities to reduce and prevent plastic leakage, in line with the Mission Ocean and Waters objectives, the EU Zero Pollution Action Plan, and the Sustainable Development Goals. The project methodology is designed to make Mediterranean

communities: a) **Scientifically credible**, through harmonised monitoring protocols aligned with MSFD TG ML, systematic data pipelines to EMODnet and DTO, and open datasets that strengthen evidence-based policymaking. b) **Operationally capable**, by validating innovative, no-harm removal technologies and equipping municipalities with operational playbooks, procurement templates, and decision-support dashboards that turn pilots into scalable, authority-owned practices. c) **Socially inclusive and circular**, by mobilising citizens, schools, and industries to co-design prevention measures, education programmes, and entrepreneurship models that transform collected plastics into usable products and new opportunities. d) **Replicable and scalable**, by developing a Mediterranean Toolbox that integrates monitoring, removal, prevention, education, and digital tools into a single platform, validated in three pilot sites and replicated through ARs. The project identifies **five methodological drivers**, conceived as an operational framework to embed and categorise the solutions: **Monitoring and Data Integration** – harmonised protocols, in-situ sensors, drones, and citizen science linked to EMODnet and DTO; **Removal Technologies** – piloting and validating innovative interception devices under real conditions with full environmental safeguards; **Prevention and Societal Uptake** – engaging municipalities, ports, fishing sectors, schools, and communities to stop plastics at source; **Circular and Educational Innovation** – enabling transformation of plastics into usable products through FabLabs, makerspaces, and training programmes; **Replication and Policy Uptake** – scaling solutions through ARs, replication roadmaps, and policy briefs that ensure long-term impact. Social justice and inclusion, climate resilience, and biodiversity protection will be considered as **cross-cutting priorities** in each solution. By integrating digital, technical, social, organisational, and governance innovations within this framework, the project builds the foundation for supporting Mediterranean regions to deliver the Mission Ocean target of a 50% reduction in plastic litter. **The project methodology is structured in four interlinked phases**, designed to ensure scientific robustness, operational feasibility, and basin-level scalability. Each phase builds on the previous one, while feedback loops between pilots, digital platforms, and stakeholder processes ensure continuous refinement and alignment with the Mission Ocean objectives. **The first phase**, from month 1 to 12, establishes the conceptual and operational foundations. Stakeholder mapping, citizen science protocols, and participatory workshops engage municipalities, schools, NGOs, and community actors to co-design prevention measures and waste-system improvements. In parallel, existing low-cost and open-source solutions for circular transformation and removal technologies are systematically mapped and assessed for applicability across diverse Mediterranean contexts. Education frameworks and training modules are drafted, drawing on established methodologies such as Universal Design for Learning and whole-school approaches. At the technical level, Toolbox architecture is defined, harmonised monitoring protocols compliant with MSFD TG ML are prepared, and baseline data pipelines are designed to ensure interoperability with EMODnet and the Digital Twin of the Ocean. This phase provides the foundation for all subsequent work by combining scientific, technical, and social perspectives. **The second phase**, from month 6 to 18, focuses on prototyping and preparation for demonstration. Co-design sessions in FabLabs and makerspaces produce prototypes of low-tech tools and circular solutions, tested together with entrepreneurs and civic groups. Pilot preparation begins in Spain (River Besòs, addressing urban–industrial pressures and community-led prevention), Greece (Remathia Stream, a sub-urban site with strong citizen science traditions and challenges in agricultural runoff), and Croatia (Neretva River, a cross-border basin with unique biodiversity and multiple municipalities involved). In each site, governance arrangements, monitoring infrastructure, and citizen programmes are set up to ensure technical readiness and community buy-in. Digital dashboards and user interfaces are tested with municipalities to confirm usability and accessibility, while citizen science initiatives are rolled out in schools and local communities to secure early ownership and engagement. **The third phase**, from month 12 to 30, represents the demonstration and validation stage. Removal technologies such as Floating Booms, Trash Traps, Bio barriers, are deployed in real river-to-delta-to-sea conditions, accompanied by life-cycle and ecotoxicity analyses to confirm no-harm safeguards. Prevention measures, including fishing-for-litter in the Besòs, waste-system fixes and citizen campaigns in Remathia, and port reception and municipal interventions along the Neretva, are implemented in cooperation with local authorities and industry. Circular entrepreneurship initiatives transform collected plastics into usable products, linking environmental goals with social innovation and local economic development. Baseline and endline monitoring are conducted to quantify plastic reductions in terms of kilograms per day, percentage flux reductions, and hotspot dynamics. Data are quality-assured and openly published within three months of acquisition. By the end of this phase, the Toolbox is populated with validated playbooks, operational guidelines, and technical documentation, providing public authorities with authority-owned, procurement-ready solutions. **The final phase**, from month 18 to 36, ensures basin-level replication and long-term uptake. An ARs scheme is launched through an open and transparent process, providing financial support of up to €100,000 per region, complemented by structured mentoring, technical assistance, and access to the Toolbox. Replication roadmaps and uptake strategies are co-developed with at least

three ARs, ensuring diffusion beyond the initial pilots. At the same time, the consortium delivers adoption roadmaps, policy briefs, and financing pathways to secure long-term sustainability. The Toolbox is released in its final form as an open digital platform fully integrated with EMODnet and DTO, hosting datasets, guides, methodologies, and decision-support dashboards. This ensures that results are not only demonstrated but institutionalised, enabling Mediterranean authorities to meet the Mission target of a 50% reduction in plastic pollution. This phased methodology reflects the project’s ambition to move beyond fragmented pilots and awareness campaigns. It integrates technical validation, citizen engagement, policy alignment, and open data into a single coherent framework, ensuring that monitoring, removal, prevention, education, and replication are all addressed in a mutually reinforcing way. In this way, the project combines TRL advancement with societal readiness and ensures that solutions are scientifically credible, socially inclusive, and operationally feasible for public authorities.

1.2.2 Specific approaches/innovations/methodologies

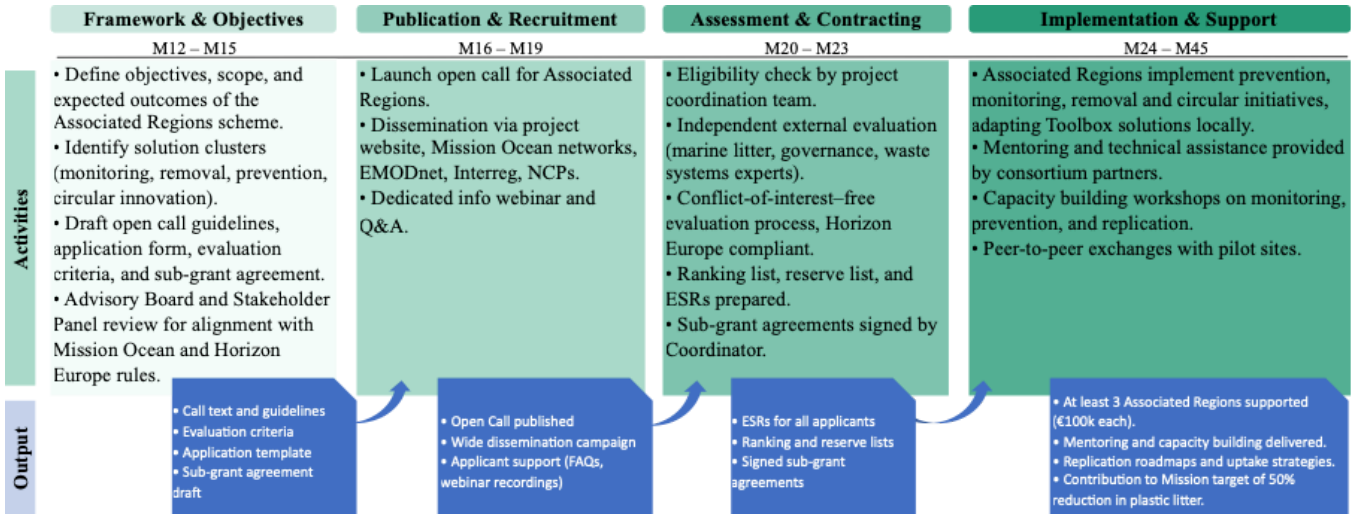
The project identifies five methodological drivers, conceived as an operational framework to embed and categorise the solutions. These drivers structure the project’s innovation pathway and demonstrate how the consortium’s unique expertise will translate into measurable progress towards the Mission Ocean target of a 50% reduction in plastic litter. **Monitoring and Data Integration** will be advanced through harmonised TG ML–compliant protocols, integrating in-situ sensors, drones, and structured citizen science campaigns. Partners such as ICCS and TechnAI bring cutting-edge expertise in environmental monitoring and digital pipelines, ensuring data flows are fully interoperable with EMODnet and the Digital Twin of the Ocean. This ensures scientific robustness, standardisation, and basin-level comparability, addressing one of the main gaps in current marine litter programmes. **Removal Technologies** will be piloted and validated under real river-to-delta-to-sea conditions, including the Besòs (urban-industrial), Remathia (sub-urban /agricultural), and Neretva (biodiversity-rich, cross-border) sites. SMEs and FabLabs will adapt and prototype interception devices such as Floating Booms, Trash Traps, Bio barriers, while scientific partners carry out full life-cycle and ecotoxicity analyses to guarantee environmental safety. This coupling of innovation with validation is a step-change from current fragmented deployments of removal devices. **Prevention and Societal Uptake** is driven by municipalities, ports, fishing sectors, schools, and community organisations. Partners like ZTKDU, Commonsplace, and Lichen bring experience in mobilising communities, while municipal authorities in Spain, Greece, and Croatia secure governance backing. Campaigns such as fishing-for-litter, behavioural nudges in schools, and municipal waste-system reforms will be embedded into authority-owned protocols and procurement templates, ensuring uptake and durability beyond awareness-raising. **Circular and Educational Innovation** leverages FabLabs and makerspaces (TransfoLAB BCN, FLZG, DecodeFabLab) to transform collected plastics into usable products, linking environmental action with social entrepreneurship. Educational programmes developed by the UZAF, University of Dubrovnik, and Colab House partners like will provide training modules for municipalities, schools, and civic actors. This integration of education with circular pilots ensures that prevention and reuse are not abstract goals but tangible local opportunities. **Replication and Policy Uptake** is secured through the ARs scheme, offering transparent cascade funding, mentoring, and access to the Toolbox. Regions not directly hosting pilots will benefit from structured roadmaps, technical assistance, and peer-to-peer exchange. Policy partners will deliver adoption briefs, procurement templates, and financing pathways, linking results to EU strategies such as the Zero Pollution Action Plan, the Marine Strategy Framework Directive, and the New European Bauhaus. Across all five drivers, **social justice and inclusion, climate resilience, and biodiversity protection** are treated as cross-cutting priorities. Vulnerable groups, including women, youth, and low-income communities, will be actively engaged in citizen science, entrepreneurship, and training activities, ensuring that the project’s solutions are not only technically sound but also socially embedded. XR-enabled **experiential learning modules**, integrating citizen science and participatory co-creation to build local capacity for plastic monitoring and circular economy practices. Knowledge transfer will be achieved through immersive simulations, hands-on data collection, and transformation of plastics into value-added products. By integrating digital, technical, social, organisational, and governance innovations, the project creates a comprehensive and replicable framework that addresses the fragmented state-of-the-art and empowers Mediterranean regions to deliver measurable and lasting reductions in plastic litter.

Methodologic al Driver	Current Gaps (State-of-the-Art)	Project Solutions & Activities	Key Partners & Strengths
1. Monitoring and Data Integration	Fragmented inland monitoring; lack of harmonised methods;	Develop TG ML–compliant protocols; deploy in-situ sensors, drones, and citizen science; ensure QA/QC;	CSIC, ICCS (monitoring & digital pipelines); ZTKDU (citizen science mobilisation)

	weak interoperability with EU systems	integrate pipelines with EMODnet and Digital Twin Ocean	
2. Removal Technologies	Devices piloted without ecological validation; few LCAs or long-term operational guidelines	Pilot low-tech, low-cost open source solutions in three river–delta–sea pilots; conduct LCA & ecotoxicity analyses; prepare O&M playbooks for municipalities	TransfoLAB BCN, SMEs, FLZG (prototyping); scientific partners (CSIC) for validation
3. Prevention and Societal Uptake	Awareness campaigns remain isolated; fishing and port sectors under-engaged; municipal reforms	Engage ports, fisheries, schools, NGOs, and municipalities; run fishing-for-litter and behavioural nudges; co-develop procurement templates and municipal waste-system improvements	ZTKDU, Commonsense, Lichen (citizen/NGO mobilisation); Municipalities in Spain, Greece, Croatia
4. Circular and Educational Innovation	Limited link between plastic collection and local economic reuse; training not embedded in governance	Develop sustainability education programmes; run circular pilots in FabLabs; enable social entrepreneurship for plastic reuse; create curricula for municipalities and schools	UZAF, University of Dubrovnik (education); TransfoLAB BCN, FLZG (circular pilots); DecodeFabLab (training design)
5. Replication and Policy Uptake	Fragmented diffusion of solutions; weak connection between pilots and policy frameworks	Launch ARs scheme (cascade funding + mentoring); co-develop replication roadmaps; prepare policy briefs and procurement/financing pathways; align with EU strategies (Zero Pollution, MSFD, NEB)	UZAF (coordination); policy partners; ARs (public authorities in Med basin)

Financial Support to Third Parties (FSTP) – ARs Scheme Following a structured open call model, RIPPLE will establish an ARs Scheme to replicate and upscale Toolbox solutions beyond the three pilot sites, ensuring basin-level impact across the Mediterranean. The scheme will provide Financial Support to Third Parties (FSTP) of up to €100,000 per Associated Region, complemented by mentoring, training, technical assistance, and early access to the Toolbox. Beneficiaries will be local and regional authorities, municipalities, river basin agencies, port authorities, or other public bodies with mandates in water, waste, or coastal management. The scheme aims to select at least three ARs (one per Mediterranean sub-basin), with the option to support additional regions if budget allows. The process will be articulated into four main steps, in line with Horizon Europe FSTP rules, and is illustrated in Figure below.

- 1. Framework & Objectives (M9–M12):** RIPPLE will prepare the open call package (D7.4), including call text, eligibility criteria, evaluation guidelines, templates for application and sub-grantee agreements. Challenges will be derived from pilot sites across four domains: monitoring, removal, prevention, and circular entrepreneurship. The Advisory Board and Stakeholder Panel (WP8) will review the framework to ensure transparency, fairness, and alignment with Mission Ocean requirements.
- 2. Publication & Recruitment (M12–M15):** The call will be published and disseminated widely via RIPPLE’s communication channels (website, social media), Mission Ocean networks, EMODnet, Interreg actors, National Contact Points, and relevant EU platforms. A dedicated webinar and Q&A session will be organised to ensure accessibility and clarify requirements. Eligible applicants will include municipalities, counties, regional authorities, and other public entities with mandates in river, delta, coastal or waste management.
- 3. Assessment & Contracting (M15–M18):** Applications will undergo a two-stage evaluation: *Eligibility check* by the Project Management Team (WP8), ensuring compliance with criteria; *Independent evaluation* by external experts in marine litter, waste management, governance, and innovation, following Horizon Europe standards of impartiality, confidentiality, and conflict-of-interest management. Evaluation Summary Reports (ESRs) will be prepared for all applicants, with a ranking and reserve list produced. Sub-grant agreements will be signed by the Coordinator on behalf of the consortium.
- 4. Implementation & Support (M18–M45):** Selected ARs will receive financial support plus intensive non-financial assistance, including: Mentoring and technical support from RIPPLE partners; Participation in training and capacity building workshops; Early access to the Toolbox, digital dashboards, and replication roadmaps; Peer-to-peer exchanges with pilot regions. Progress will be monitored and reported regularly to ensure accountability, transparency, and uptake of results.



1.2.3 Pilot activities

The project has selected **three complementary Mediterranean pilots** — the Besòs River (Spain), the Neretva River (Croatia), and the Remathia Stream (Greece) — representing distinct hydrological, socio-ecological, and governance contexts of plastic pollution. **Besòs**: highly urbanised, flood-prone river discharging plastics directly into a metropolitan coast. **Neretva**: large cross-border river, among the most polluted in the Adriatic, linking inland waste mismanagement to marine litter hotspots in a protected delta. **Remathia**: a smaller sub-urban stream, showing how even minor waterways act as pathways for plastics, especially when stormwater and drainage systems are insufficient. Together they provide a gradient of scales and conditions (urban–industrial, deltaic–cross-border, and sub-urban –coastal) that are representative of Mediterranean challenges. This ensures that solutions tested are transferable and scalable to diverse contexts across the basin.

Pilot 1: River Besòs (Spain) The River Besòs is one of the most emblematic examples of an urbanised Mediterranean river. Flowing for only 18 kilometres through the Barcelona metropolitan area before reaching the sea, it is heavily affected by urban runoff, combined sewer overflows, and episodic floods that carry significant amounts of litter downstream. Recent studies estimate that the Besòs discharges up to 1.7 tonnes of plastic annually, accompanied by billions of microplastic particles. Surveys show a dominance of single-use plastics such as bags, bottles, and packaging, reflecting the consumption and waste patterns of the metropolitan area. The riverbanks and estuary also accumulate plastic debris during flood events, aggravating local flood risks and environmental degradation. Several initiatives have already tackled the Besòs, such as the monitoring campaigns of the Plastic Pirates citizen science programme, local clean-up actions by NGOs like Pescadors de Plàstics, and technical studies by the Agència Catalana de l’Aigua (ACA). However, these efforts remain fragmented and lack systematic harmonisation with European frameworks such as the MSFD Technical Group on Marine Litter (TG ML). The Besòs pilot will build on these initiatives by applying TG ML–compliant monitoring protocols, integrating data flows into EMODnet, and ensuring interoperability with the Digital Twin of the Ocean. In terms of project objectives, the Besòs directly contributes to SO1 (harmonised monitoring and data systems) through the establishment of real-time monitoring with drones, in-situ sensors, and citizen science protocols, and to SO4 (demonstration of sustainable removal technologies) by deploying and validating low-tech low-cost and open source solutions in a flood-prone river. It will also address SO3 (prevention and uptake measures) by working with local schools and NGOs, municipalities, and the waste operator to implement awareness campaigns and waste-system improvements. Finally, through collaboration with the Consorci Besòs-Tordera and the Barcelona municipalities, the pilot will ensure that procurement-ready playbooks and operational guidelines are embedded in local governance, directly linking to TO 4.3 on operational playbooks. The Besòs pilot will be coordinated by TeamLabs, part of Mondragon Universitatea with a mandate from Ajuntament de Barcelona, the main public authority. TeamLabs is the organisation leading Barcelona Circular, the new circular innovation lab, whose main purpose is to promote the circular economy in the city. It is based in the proximity of the Besos river, within el Bon Pastor neighbourhood, and is one of the innovation hubs prioritized in Barcelona’s city council economic strategy. Technical expertise will come from TransfoLAB BCN, TechnAI and CSIC, while social engagement will be facilitated by Lichen through local schools and NGOs. Key private stakeholders include the metropolitan waste operator (AMB) and companies involved in drainage and flood protection. The co-creation mechanism involves three layers:

- Institutional roundtables, where the project partners and relevant stakeholders will validate monitoring protocols, agree on removal technologies, and align prevention actions with existing waste-management plans.
- Community co-creation sessions, which bring together citizens, NGOs (e.g., Pescadors de Plàstics), and schools to shape awareness campaigns and test citizen science tools.
- Pilot testing workshops in FabLabs with entrepreneurs, ensuring that circular solutions and playbooks are co-designed with both civic and private actors.

This layered governance ensures that technical validation (SO1–SO2) is directly linked with societal uptake (SO3), and that outputs such as procurement templates (TO 1.3) are grounded in the operational reality of municipalities.

Pilot 2: River Neretva (Croatia and Bosnia and Herzegovina) The Neretva River is one of the largest rivers in the Adriatic and has been identified as one of the most plastic-polluted rivers globally, contributing an estimated 2.8% of the plastic load into the entire Adriatic Sea. Flowing 225 kilometres from Bosnia and Herzegovina into Croatia, the Neretva enters the Adriatic through a fertile delta that is both ecologically sensitive and economically vital. The delta is a Ramsar site, supporting agriculture, fisheries, and tourism, yet it suffers from severe plastic accumulation on its banks, beaches, and wetlands. Floods exacerbate the problem by flushing mismanaged waste from upstream municipalities into the river. Field surveys in the delta indicate that more than 45% of collected litter consists of single-use plastics, highlighting systemic weaknesses in waste management. The Neretva region has been the focus of multiple EU-funded projects, such as SEADRION and MARLESS, which assessed marine litter in the Adriatic, and cross-border cooperation initiatives led by the Dubrovnik-Neretva County and the Regional Development Agency DUNEA. These provide a solid foundation but have not yet advanced into operational removal, systemic prevention, or authority-owned replication. The Neretva pilot will bridge this gap by deploying removal technologies under real river-to-sea conditions, coupled with full life-cycle assessments and ecotoxicity validation to ensure environmental safety. This pilot links strongly to SO2 (demonstrating safe removal technologies) through the testing of low-tech low-cost and open source solutions, and to SO5 (demonstrating the Toolbox) by delivering quantified reductions in plastic fluxes. It also advances SO3 (prevention and uptake) via the introduction of fishing-for-litter schemes, improved port reception systems in Ploče, and municipal waste-system reforms in Metković and Opuzen. In addition, SO6 (capacity building and education) will be addressed through school programmes, awareness campaigns, and circular entrepreneurship initiatives that enable local communities to reuse collected plastics. This pilot is strategically important because of its cross-border character: the Neretva demonstrates how upstream waste mismanagement in one country directly affects downstream ecosystems in another, underlining the need for basin-level coordination and replication. The Neretva pilot will be anchored in the Dubrovnik-Neretva County Authority as the lead public body, supported by the Regional Development Agency DUNEA for cross-border cooperation. Municipalities of Metković, Opuzen, and Ploče will host site-level activities. On the BiH side, links with the University of Mostar and Agency for Watershed of Adriatic Sea Mostar will be activated for upstream engagement. Scientific and technical partners include the University of Dubrovnik, UZAF, and Fablab Zagreb. Fisheries cooperatives, port authorities in Ploče, and local waste operators are key sectoral actors. The co-creation mechanism here is designed around the delta governance model:

- County-led Steering Committee, integrating municipalities, ports, and fisheries, meets biannually to prioritise interventions, validate monitoring data, and adopt playbooks.
- Cross-border coordination workshops facilitated by DUNEA, where Croatian and BiH stakeholders jointly review flux monitoring results and align prevention measures, ensuring transboundary cooperation.
- School and community campaigns run by ZTKDU and NGOs in the Neretva valley, focusing on behavioural change and linking citizen-collected data to official monitoring.

This design ensures that removal technologies (SO2) and monitoring protocols (SO1) are embedded in real governance settings, while prevention measures (SO3) are co-owned by the fishing and port sectors that directly influence plastic flows.

Pilot 3: Remathia Stream (Greece)

The Remathia Stream in Chalandri, a suburban municipality in the Attica metropolitan area, represents a smaller yet highly impactful pathway of plastic pollution. Although only a few kilometres long, the stream connects into Kifissos river system and carries considerable amounts of plastic litter during rainfall events, reflecting weaknesses in urban drainage, stormwater management, and waste collection systems. During dry season the plastics accumulate along the stream banks, at outfalls, and in nearby public spaces, directly affecting the local community and eventually reaching the **Saronic Gulf**, a semi-enclosed basin highly sensitive to pollution. Unlike larger rivers, small suburban streams such as Remathia have often been neglected in marine litter policies, despite their importance as diffuse entry points of plastics into the sea. Local authorities, including the **Municipality of**

Chalandri, have expressed strong interest in addressing this gap. The Municipality of Chalandri's management study for Remathia, approved by the Forestry Directorate in 2024, aims to safeguard the rare ecosystem and landscape from building pressures, vegetation aging, and natural disasters, within the framework of what is provided for the protection of the stream in the Presidential Decree 659 D/1995 (Government Gazette). Previous EU initiatives such as LIFE DEBAG, which targeted plastic bag reduction in Greece, and the participation of local schools in the Plastic Pirates campaign, provide an entry point for mobilising citizens and young people. The Municipality of Chalandri also implemented the CULTURAL H.ID.RA.NT EUI Project (2021–2024), focusing on the Hadrianic Aqueduct as a vehicle for showcasing the city's cultural identity and heritage. The project combined extensive participatory community processes, completing more than a 100 workshops, with cultural and environmental actions, emphasizing the role of water as both a natural resource and a unifying element for social engagement and urban sustainability. Within the project, building upon the existing experience, Remathia will contribute to SO1 (harmonised monitoring) through the implementation of citizen science and low-cost monitoring adapted to small streams, and to SO2 (removal technologies) by testing low-tech devices such as floating traps and smart bins that are appropriate for smaller waterways. It will also advance SO3 (prevention and uptake) by running targeted school programmes and behavioural nudges in local communities. Importantly, Remathia demonstrates the scalability of solutions to suburban contexts that exist across the Mediterranean, making it essential for ensuring the transferability and replication potential of the Toolbox. The Remathia Stream pilot is led by KYVELI MAE mandated by the Municipality of Chalandri to perform developmental activities on their behalf, with regulatory oversight from the municipality. Technical expertise is provided by Decode Fab Lab, ICCS and CoLab House, while community engagement is supported by Commonsense, CoLab House and local schools. Businesses located near the stream, including small retailers and service providers, are also identified as stakeholders due to their role in generating and managing urban litter. The co-creation mechanism involves:

- Municipal stakeholder forums, chaired by the Municipality of Chalandri and KYVELI MAE, aligning monitoring data and stream interventions with local environmental and urban policies.
- Co-design sessions with the creative community and citizens organised by CoLab House, Decode Fab Lab and Commonsense, focusing on technical solutions evaluation and testing, behavioural nudges, and developing school educational and engagement programmes.
- Authority–community joint reviews hosted by KYVELI MAE, CoLab House and Commonsense where monitoring results (baseline and endline) are translated into actionable measures and validated by both technical experts and citizens.

This ensures that smaller, suburban waterways—often neglected in marine litter strategies—become active sites of governance innovation, directly contributing to prevention (SO3), monitoring (SO1), and demonstration of scalable solutions (SO4).

1.2.4 Links with existing national and international projects

Project	Focus	Relation to our project
Mission Ocean & Waters Implementation Platform(<i>mandatory</i>)	Coordination, monitoring, impact framework for Mission Ocean	Ensures Toolbox design and replication are fully aligned with Mission governance and monitoring targets.
Plastic Pirates (<i>mandatory</i>)	Citizen science with schools and youth to collect plastic data	Direct input for WP1 citizen science protocols and WP3 education modules, ensuring comparability and wide engagement.
WATER4ALL Partnership(<i>mandatory</i>)	Water management, governance, and research infrastructures	Provides synergies for basin-level governance and integration of our monitoring datasets.
INSPIRE / REMEDIES(Horizon Europe)	Innovative prevention and removal technologies for plastics	Complement our pilots in WP6 by linking tested devices and behavioural interventions.
CLAIM (H2020)	Marine litter monitoring, modelling, and removal systems	Extended in our project with LCA, ecotoxicity, and deployment in Mediterranean hotspots.
CLEANATLANTIC (Interreg)	Atlantic-wide litter monitoring and awareness	Baseline methodologies transferred to Mediterranean pilots for harmonised monitoring.
BLUEISLANDS (Interreg MED)	Seasonal plastic fluxes in Mediterranean islands	Provides insights on tourism-driven litter flows relevant to pilot sites (Croatia, Greece).

Making Sense (H2020)	Open-source sensing and digital maker practices	Inspires WP2/WP5 approaches for low-cost monitoring and open digital tools.
dRural (H2020)	Digital platforms and marketplaces for rural areas	Relevant for Toolbox development and WP7 dissemination to local authorities.

1.2.5 Transdisciplinary Integration (including social sciences and humanities)

The consortium combines a broad spectrum of disciplines and actors, ensuring full integration of STEM and SSH throughout the project. Universities and research institutes (UZAF, University of Dubrovnik, CSIC, ICCS) provide expertise in marine and freshwater ecology, environmental monitoring, spatial planning, architecture and digital innovation. Civil society organisations (Lichen, Commonsense, CoLab House, ZTKDU, Institute POI,) lead participatory design, citizen science and education, while creative and maker organisations (TransfoLAB BCN, Decode Fab Lab, FLZG) bring circular design and low-cost innovation. SMEs and digital partners (Technai, ICCS) develop monitoring, data pipelines and dashboards interoperable with EMODnet and the Digital Twin Ocean. Local and regional authorities (TeamsLab mandated by the Ajuntament de Barcelona, KYVELI MAE mandated by the Municipality of Chalandri, Dubrovnik-Neretva Developing Agency on behalf of Dubrovnik Neretva County) act as site owners and ensure governance relevance, replication and uptake. The project integrates SSH and STEM at every stage: monitoring protocols are co-designed with citizen science and governance insights; removal technologies are validated against ecological and social acceptance criteria; prevention measures combine behavioural insights, education and sectoral engagement; replication plans embed policy, financing and community perspectives. SSH expertise covers sociology, anthropology, behavioural economics, environmental governance and education sciences, ensuring that solutions address cultural, social and institutional dimensions alongside technical ones. Transdisciplinary methods such as participatory planning, design thinking, system dynamics and citizen science protocols ensure that solutions are robust, user-centred and adaptable across contexts. The UZAF, as coordinator, embodies this integration by linking architecture, urbanism, landscape and environmental sciences with technical and social innovation. The pilots further reinforce transdisciplinarity by combining urban (Barcelona), sub-urban (Greece) and riverine–fisheries (Croatia) contexts, ensuring solutions are tested in diverse socio-ecological systems. Through cascade funding, ARs gain mentoring, training and access to the Toolbox, extending the integration of disciplines and actors beyond the consortium. This approach guarantees that the Toolbox is not only technically and environmentally sound, but also socially inclusive, politically relevant and institutionally embedded, creating long-term value beyond the lifetime of the project. In line with the European Commission's First Monitoring Report on SSH Integration in Horizon Europe (2025), which highlights that SSH partners are often underrepresented in leadership roles, our consortium ensures that SSH actors lead key work packages on participation, education, and uptake, fully embedding social innovation alongside technical innovation.

1.2.6 Gender Dimension

The project recognises that plastic pollution is not gender-neutral, as women and men interact differently with plastics through household consumption, waste sorting, education and civic initiatives, while fisheries and port sectors remain predominantly male. These dynamics shape both exposure to plastic litter and opportunities for prevention. In line with *Gendered Innovations 2: How Inclusive Analysis Contributes to Research and Innovation* (European Commission, 2020), sex- and gender-based analysis is integrated into the research content to strengthen excellence, creativity and societal relevance. In co-creation and citizen science activities (WP2, WP6) the project will ensure balanced participation and, where relevant, collect gender-disaggregated data such as in litter collection, classification and perception surveys. Educational programmes and awareness campaigns (WP4) will be designed with gender-sensitive narratives that reach diverse audiences and avoid stereotypes, while the Toolbox (WP5) will include recommendations for public authorities on how to integrate gender perspectives into monitoring, prevention and communication. Replication through ARs will transfer this approach locally, ensuring that differences in community roles and needs are considered in uptake. By embedding gender and intersectionality into methods, participation and outputs, the project addresses a known weakness in Horizon proposals, namely insufficient integration of gender into research content, and ensures that the Toolbox is not only technically and environmentally robust but also socially inclusive, equitable and aligned with Horizon Europe's priorities for gender-sensitive R&I. Gender balance is a key strength of the consortium, with 55% of the researchers involved in the project being women.

1.2.7 Open science practices

In line with the EU and Horizon Europe Open Science principles, which promote openness, transparency, and inclusivity in research and innovation, the project will adopt a Quadruple Helix innovation model that brings

together academia, industry, government, and civil society in every stage of the process. This approach will ensure that all project outcomes are demand-driven, credible, and directly relevant to society. A Multi-Actor Approach will be applied throughout the project, integrating municipalities, river basin authorities, ports, fisheries, schools, NGOs, SMEs, and citizens into the co-design of monitoring, prevention, removal, and circular transformation solutions. In this way, the principles of Responsible Research and Innovation will guide all activities, making the research process fairer, more inclusive, and more impactful. All research outputs, including datasets, protocols, digital tools, models, and reports, will be openly accessible and compliant with the FAIR principles, ensuring that they are findable, accessible, interoperable, and reusable. The consortium will guarantee that results are deposited in trusted repositories such as Zenodo, Figshare, EMODnet, and OpenAIRE, while also ensuring interoperability with the EU Digital Twin Ocean. Publications will be made openly available either via Gold or Green Open Access, in line with Horizon Europe requirements, and all manuscripts will also be self-archived in institutional repositories and on the project website. A Data Manager within WP8 will oversee documentation, publication, and compliance with FAIR principles, ensuring alignment with the European Open Science Cloud objectives that require publicly funded research to make its results as openly available as possible and to ensure that research data produced in Europe is FAIR by design. The Open Science strategy is fully embedded in the methodology of the project. In all three pilot sites—Remathia, Besòs, and Neretva—local stakeholders such as municipalities, waste companies, fisheries, schools, NGOs, and industrial actors will actively co-create solutions. Citizen science protocols, aligned with the MSFD TG ML guidelines, will enable schools and communities to collect data on plastics in rivers and deltas. These data will be validated and uploaded to EMODnet within three months of acquisition, guaranteeing both timeliness and quality. Vulnerable groups, including youth, women, and low-income communities, will be actively engaged to democratise the scientific process and maximise societal impact. All monitoring data collected through in-situ sensors, drones, citizen science, and laboratory analyses will be documented, quality-controlled, and published openly, following common standards on metadata and QA/QC protocols to ensure comparability and reuse. Joint exploitation with existing European datasets, including those generated by initiatives such as Plastic Pirates, EMODnet, and WATER4ALL, will be pursued to enhance alignment and synergies. The project will produce a minimum of ten scientific publications in peer-reviewed Open Access journals or on Open Research Europe, under Creative Commons licences such as CC-BY or CC0. In addition, preprints will be deposited in open platforms before peer review, ensuring that knowledge is shared early and transparently. This principle of early and open sharing will also apply to datasets and protocols, which will be made available in the first stages of the project to secure immediate reusability and transparency. Where possible, the project will encourage open peer review, and stakeholder feedback will be systematically integrated into reports and guidance documents such as the Toolbox, playbooks, and procurement templates. Reproducibility will be supported by publishing code, algorithms, and digital tools in open-source repositories such as GitHub and GitLab. Open Lab Notebooks will be used where appropriate to foster collaboration with the wider research community, and clear guidelines will be provided to enable replication of results and transferability of solutions to other basins. By embedding these practices across all work packages, the project will ensure that its outcomes—including harmonised monitoring protocols, validated removal technologies, citizen science methodologies, digital dashboards, and circular transformation practices—are not only scientifically robust but also open, transparent, and ready for widespread uptake.

1.2.8 Data management and management of research outputs

The research data and outputs generated in this project will be managed responsibly, fully in line with the FAIR (Findable, Accessible, Interoperable, Reusable) principles and the requirements of Article 17 of the Annotated Model Grant Agreement. An initial Data Management Plan (DMP) will be prepared by Month 6 under WP8, describing in detail how all data generated will be collected, processed, shared, and preserved by project partners. The DMP will also address ethical safeguards, intellectual property rights, confidentiality, and publication provisions as defined in the Consortium Agreement. Updates to the DMP will be made regularly to reflect changes in consortium policies, the identification of new innovation potential, or new decisions on data exploitation and protection. The project will generate and reuse multiple types of data, reflecting its interdisciplinary nature. These will include numerical and tabular data from monitoring activities; geospatial data from drones and in-situ sensors; textual and qualitative data from participatory workshops and citizen science campaigns; laboratory data from LCA and ecotoxicity assessments; image and video data from pilot deployments; and metadata and documentation linked to protocols, playbooks, and training modules. To ensure findability, all datasets and research outputs will be deposited and described in trusted disciplinary, institutional, or multi-disciplinary repositories that guarantee long-term preservation and persistent unique identifiers (DOI, Handle, Zenodo, etc.). Chosen repositories will be

registered in the re3data directory and harvested by OpenAIRE, guaranteeing visibility within the European Open Science Cloud (EOSC). Accessibility will be maximised by making research data underlying scientific publications and public deliverables openly available immediately upon publication, in line with Horizon Europe Open Access requirements. Other project datasets will be deposited by the end of the project. Restrictions to access will only be applied when strictly necessary, for example due to privacy, copyright, confidentiality, or exploitation concerns. Such restrictions, together with the justification, will be documented in the DMP. Interoperability will be ensured by depositing data with standard descriptive metadata, such as Dublin Core and DataCite Metadata Schema, and by using repositories that provide OAI-PMH interoperability. This will ensure indexing, discoverability, and comparability with other datasets, especially within EMODnet and the Digital Twin Ocean (DTO). Reusability will be supported by adopting open licences such as Creative Commons Attribution (CC-BY) or Creative Commons Public Domain Dedication (CC0). Where specific data require different licensing, this will be clearly explained in the DMP. All deposited data will be accompanied by relevant documentation — including codebooks, protocols, and methodological notes — to enable validation, interpretation, and reuse by external actors. Open-source repositories (e.g., GitHub, GitLab) will be used for sharing software, models, and algorithms developed during the project, ensuring transparency and reproducibility. Costs related to data collection, cleaning, anonymisation, documentation, conversion into open formats, and long-term preservation are included in the estimated partner budgets. Each partner will remain responsible for the quality, organisation, and secure storage of the data they generate, while the coordinator will oversee implementation of the DMP. A dedicated Data Manager will be appointed within WP8 to ensure usability, accountability, and quality of all project data, and to maximise its scientific and societal value. Finally, capacity-building activities will be organised both internally and externally to raise awareness about good data management practices. Training sessions will support partners, local authorities, and associated regions in understanding how to collect, curate, and reuse open data. #§CON-MET-CM\$# #§COM-PL-CP\$# #§REL-EVA-RE\$#

2. Impact #@IMP-ACT-IA@#

2.1 Project's pathways towards impact

2.1.1 Contribution towards outcomes of the Call

Expected Outcome	Project Contribution
Public authorities and relevant stakeholders have access to a toolbox to address marine plastic litter from land-to-river-to-sea	The project co-designs and delivers a Mediterranean Toolbox integrating (a) harmonised monitoring protocols and open data flows, (b) validated removal technologies, and (c) societal and sectoral engagement measures. The Toolbox is co-developed with municipalities and water managers in Spain, Greece and Croatia, and validated through use-cases before replication in ARs.
Reduced plastic litter pollution in European rivers and seas at basin level	Three pilot sites demonstrate measurable reductions in plastic fluxes to the sea through combined interventions: MSFD-compliant monitoring, sustainable removal devices, and prevention measures with citizens, schools, and the fishing sector. Results are quantified against baselines (kg/day, % reduction).
Accelerated uptake of innovative solutions to prevent and remove litter	The project pilots innovative removal devices and provides operational playbooks, maintenance models and cost-effectiveness assessments for public authorities. Preventive measures (waste-system improvements, fishing-for-litter schemes, school programmes) are co-designed with users, ensuring adoption and replication.
Enhanced knowledge on marine litter in line with MSFD TG Marine Litter guidelines	Monitoring activities generate harmonised datasets across compartments (surface, bank, underwater, micro/macro plastics). Citizen science protocols are aligned with TG ML. Datasets are quality controlled and openly shared via EMODnet, enhancing comparability and policy relevance.
Measurable contributions to Mission Ocean and EU Zero Pollution targets (50% reduction of plastic in the sea)	Project pilots deliver verified flux reductions in the Mediterranean basin. Replication with ≥ 3 ARs via cascade funding scales the approach, directly contributing to Mission Ocean targets and EU Zero Pollution objectives.
Contribution to the Mission's Digital Ocean and Knowledge system	Monitoring and modelling data are published to EMODnet, with interoperability ensured for the Digital Twin Ocean (DTO). Dashboards and APIs developed in WP7 enable authorities to visualise, reuse and integrate data into planning and policy. Synergies with Plastic Pirates, WATER4ALL and Interreg ensure

continuity and European knowledge sharing.
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2.1.2 Contribution to Relevant Policies

RIPPLE is designed to provide measurable and traceable contributions to Mission Ocean. Monitoring datasets from pilots and citizen science will be harmonised with TG ML protocols and deposited in EMODnet, ensuring interoperability with the Digital Twin Ocean (DTO). Annual Mission Ocean monitoring reports will integrate RIPPLE's quantified outcomes (flux reduction, removal efficiency, citizen engagement), directly feeding into Mission tracking of progress towards the 50% plastic reduction target. The Toolbox, dashboards, and replication roadmaps will be channelled to the Mission Implementation Platform for validation and scaling. Policy briefs will align with the Zero Pollution Action Plan, MSFD, and regional marine strategies, ensuring that RIPPLE results inform EU and Member State decision-making. The Associated Regions scheme operationalises basin-level replication in line with Mission Ocean objectives, embedding Toolbox use in additional authorities beyond the pilots.

Policy / Strategy	How the Project Contributes
European Green Deal	Contributes to the Green Deal ambition of zero pollution and climate neutrality by reducing plastic pollution across rivers and seas, improving water quality, and enabling circular use of resources through prevention and reuse of plastics.
EU Zero Pollution Action Plan (2021)	Directly targets the Zero Pollution goal of reducing plastics in the sea by at least 50% by 2030. Demonstrates effective monitoring, removal, and prevention measures at basin scale and packages them into a replicable Toolbox for public authorities.
EU Biodiversity Strategy for 2030	Reduces pressures on aquatic and coastal ecosystems by cutting litter inputs, protecting marine and freshwater biodiversity, and supporting restoration of habitats impacted by plastics.
EU Circular Economy Action Plan (2020)	Promotes prevention and circularity through reuse and transformation of collected plastics in community labs, supporting entrepreneurship and local green economies.
EU Climate Adaptation Strategy (2021)	Strengthens resilience of coastal and riverine regions by reducing plastic-related stresses on ecosystems, improving ecosystem services, and integrating adaptive waste management solutions.
Mission “Restore our Ocean and Waters by 2030”	Directly aligned with Mission Ocean's target of 50% less plastic litter in the sea, implementing the land–river–sea approach in the Mediterranean basin, delivering a Toolbox for public authorities, and scaling impact through ARs.
EU Zero Pollution Monitoring and Outlook (linked to MSFD)	Embeds the latest MSFD TG Marine Litter guidelines into monitoring, ensures open data flows to EMODnet, and enhances knowledge for Europe-wide assessments of aquatic litter.
Marine Strategy Framework Directive (MSFD)	Provides harmonised, MSFD-compliant data on marine litter sources, pathways, and impacts, enabling Member States to meet obligations under Descriptor 10 (Marine Litter).
Digital Europe / EU Digital Twin Ocean	Contributes datasets and services interoperable with the Digital Twin Ocean, ensuring continuity and legacy of monitoring and forecasting tools beyond the project.
New European Bauhaus (NEB)	Embeds NEB principles of sustainability, inclusivity, and beauty by linking technical solutions with participatory design, education, and local creativity in prevention and circular reuse of plastics.
EU Gender Equality Strategy 2020–2025	Integrates gender-sensitive approaches in citizen science, co-creation, and education, ensuring balanced participation, sex-disaggregated data collection, and equitable access to solutions.
Relevant Regional Strategies (Mediterranean Action Plan, Barcelona Convention)	Supports regional cooperation by addressing marine litter in the Mediterranean basin, aligning with existing frameworks and creating synergies with Interreg and Water4All projects.

2.1.3 Assessment of Impacts

Impact Category	Key Impacts (Short-Term)	Long-Term Impacts
Scientific &	Advancing knowledge of inland sources, pathways,	Establishment of basin-wide evidence

Knowledge Impact	and hotspots of plastic litter. Application of harmonised MSFD TG ML protocols in rivers and deltas. Generation of open datasets through EMODnet and DTO-ready services.	base for monitoring and policy. Contribution to long-term EU standards and global leadership in marine litter science.
Policy & Governance Impact	Delivery of a co-designed Toolbox for municipalities, counties, and water agencies. Demonstration in three pilots and replication through ARs. Capacity building for public authorities.	Strengthened governance capacity across Europe. Uptake of Toolbox solutions in policy frameworks, supporting MSFD, Zero Pollution and Mission Ocean targets.
Technological & Innovation Impact	Demonstration of removal technologies (Floating Booms, Trash Traps, Bio barriers,) in real conditions with ecological validation. Development of digital dashboards and APIs interoperable with EMODnet and DTO.	Widespread adoption of sustainable removal and monitoring technologies. Establishment of interoperable, no-harm solutions as standard practice for European waters.
Societal & Behavioural Impact	Engagement of citizens, schools, NGOs, and fishing sector through co-creation, education, and campaigns. Collection of sex-disaggregated data in citizen science. Implementation of prevention measures at community level.	Long-term behavioural change towards prevention and circularity. Greater social equity and inclusion through gender-sensitive approaches. Empowered local communities across the basin.
Environmental & Ecosystem Impact	Demonstrated reduction of plastic litter fluxes at pilot sites (kg/day, % flux reduction). Improved water quality and reduced pressures on aquatic and coastal ecosystems.	Contribution to Mission Ocean and EU Zero Pollution targets of 50% less plastic in the sea. Protection of biodiversity and improved ecosystem services for adaptation and resilience.
Economic & Market Impact	Stimulation of circular economy entrepreneurship through reuse of plastics in labs and SMEs. Engagement of fishing communities in litter collection schemes.	Growth of sustainable industries in monitoring, removal, and circular plastics. Reduced economic losses from marine litter in fisheries, ports, and tourism. Long-term cost savings for public auth.

2.1.4 Key Performance Indicators

All KPIs will be measured against a 2025 baseline established at the three pilot sites (Spain, Greece, and Croatia). Baseline data will be collected during Months 1–6 using harmonised MSFD TG Marine Litter protocols, complemented by citizen science surveys, drone imagery, and in-situ sensors to ensure comparability across sites. Progress will be monitored annually, with data collected by local scientific partners (CSIC, UZAF/UNIDU) and verified through QA/QC checks coordinated under WP2 and WP6. Prevention and engagement KPIs will be tracked by municipalities, NGOs, and schools through attendance records, disaggregated surveys, and adoption reports. Digital and policy-related KPIs will be monitored by WP7 (via data dashboards and DTO interoperability testing) and WP8 (through policy briefs, financing pathways, and adoption agreements). A consolidated monitoring report will be produced annually by the coordinator to guarantee methodological consistency across pilots and ARs. All validated datasets will be uploaded to EMODnet within three months of acquisition and tested for DTO interoperability. Final endline measurements at Month 48 will enable robust before/after comparisons and quantification of overall project impact.

SOs	Key Result (R)	KPI	SMART Justification	Linked Call EOs
SO1	Citizen science protocols, awareness campaigns, municipal procurement templates	≥ 20 schools engaged; ≥ 6 municipalities adopting templates	S: Concrete actions (protocols, campaigns, procurement). M: Quantified by # schools, # municipalities. A: Feasible with pilot regions & NGO partners. R: Directly supports prevention at source. T: Protocols by M18; adoption by M30.	Public authorities and stakeholders have access to toolbox elements for prevention; Accelerated uptake of solutions
SO2	Training programmes, FabLab pilots,	≥ 5 modules; ≥ 300 trained; ≥ 30 prototypes	S: Focused on education & reuse. M: Tracked by # modules, participants, prototypes. A: Achievable with FabLabs & universities.	Uptake of innovative solutions;

	circular prototypes		R: Builds capacity & drives behaviour change. T: Modules by M12; prototypes by M30.	Behavioural changes embedded
SO3	TG ML-compliant monitoring protocols; open datasets to EMODnet, DTO	≥ 20 datasets; ≥ 10 monitoring stations; 100% TG ML compliance	S: Clear scientific protocols. M: Quantifiable datasets & stations. A: Achievable via RTO & ICT partners. R: Fills monitoring gap in inland waters. T: Baseline by M12; datasets uploaded ≤ 3 months.	Enhanced knowledge on marine litter; Contribution to Mission Digital Ocean
SO4	Pilots of collection technology; LCA & ecotoxicity reports	≥ 5 devices; ≥ 500 kg/day removed; 3 LCA/ecotoxicity reports	S: Specific technologies validated. M: Quantified by devices deployed & kg/day removed. A: Feasible with SME prototypes at TRL5–6. R: Ensures environmental safety. T: Deployment by M18; validation by M30.	Reduction of litter in rivers/seas; No-harm sustainable removal
SO5	Integrated Toolbox tested in Spain, Greece, Croatia	≥ 30% flux reduction; ≥ 10 playbooks/guides	S: Toolbox as operational product. M: Measured by % reduction & # guides. A: Achievable with pilots + monitoring. R: Demonstrates end-to-end solution. T: Results validated by M36.	Toolbox validated in 3 use-cases; Basin-scale approach demonstrated
SO6	ARs scheme; replication roadmaps	≥ 3 ARs mentored; ≥ 3 roadmaps	S: Transparent FSTP scheme. M: # regions supported & roadmaps delivered. A: Feasible with €100k per region. R: Ensures diffusion beyond pilots. T: Selection by M12; roadmaps by M30.	Replication in ARs; Basin-wide scaling
SO7	Policy briefs, financing pathways, exploitation strategy	≥ 6 policy briefs; ≥ 3 financing pathways; ≥ 200 Toolbox users post-project	S: Policy deliverables defined. M: Measured by # briefs, pathways, Toolbox users. A: Feasible with partners' policy expertise. R: Secures uptake beyond project. T: Deliverables finalised by M36.	Policy impact; Contribution to EU Zero Pollution & Mission Ocean targets

Scale and Significance of Impacts

Component	Scale & Target	Significance & Expected Outcome	KPIs & Quantified Impact (vs 2025 baseline)
Pilots & ARs	3 pilots (Spain, Greece, Croatia) + ≥3 ARs	Demonstration of Toolbox in diverse contexts (urban delta, coastal river, fisheries) with replication in additional authorities.	≥25% plastic flux reduction at pilot hotspots; ≥6 authorities directly applying Toolbox; ≥3 ARs with replication plans.
Public Authorities Engaged	≥10 authorities (3 pilots, 3 ARs, plus local water/port agencies)	Strengthened governance capacity and alignment with MSFD/Zero Pollution targets.	≥10 authorities using monitoring/removal/prevention modules; 2+ policy endorsements referencing Toolbox.
Citizen & Stakeholder Engagement	≥3,000 citizens and ≥30 schools across pilots; NGOs, fishing communities, waste operators	Builds societal ownership, awareness and behavioural change towards prevention and circularity.	3,000+ citizens involved in citizen science; 100,000 people reached by awareness campaigns; ≥10 tonnes plastic collected via fishing-for-litter.
Technological Validation	3–5 removal solutions (Floating Booms, Trash Traps, Bio barriers,) + digital dashboards, APIs	Validation of sustainable, cost-effective, no-harm removal and decision-support tools.	≥2 validated removal solutions per pilot; €/kg removed benchmarked ≤ EU projects; 100% datasets EMODnet/DTO-compliant.
Scientific	Universities and	New knowledge on plastic	≥15 monitoring stations; ≥90% TG ML

Contribution	research institutes in 3 countries + citizen science	litter pathways and hotspots using harmonised TG ML protocols.	compliance; ≥3 peer-reviewed papers; 100% datasets FAIR and open in EMODnet.
Circular Economy & Market Impact	SMEs in Spain, Greece, Croatia; fishing sector	Creates entrepreneurship opportunities through reuse of plastics; engages fisheries in collection.	≥5 SMEs/FabLabs piloting circular reuse; ≥3 community products created; reduced litter costs for ports/fisheries.
Policy & Replication Impact	Regional authorities and EU Mission platforms	Scaling Toolbox beyond pilots; contributing to Mission Ocean's 50% reduction target.	≥3 policy briefs; Toolbox referenced in EU or regional guidance; contribution of ~100 tonnes plastic prevented from reaching the sea during project lifetime (3 pilots + 3 ARs).

2.1.5 Requirements and Potential Barriers for Achievement

Barrier / Requirement	Potential Challenge (Beyond Project Scope)	Project Response & Mitigation Measures
Political & Governance	Shifts in political priorities or limited support from public authorities for long-term uptake of Toolbox measures.	Early engagement of municipalities, counties, and water agencies in co-design. Letters of Intent secured before deployment. Alignment with EU Mission Ocean targets provides leverage for continuity beyond project lifetime.
Economic & Financial	Limited financial capacity of local authorities to maintain monitoring or removal technologies after project funding ends.	Provision of cost-effectiveness studies, O&M playbooks, and training. Exploration of alternative funding (EU cohesion funds, EIB/EBRD green financing, polluter-pays schemes).
Social & Behavioural	Low citizen participation or resistance to prevention measures (e.g. waste separation, fishing-for-litter).	Strong participatory processes with schools, NGOs, and fishing communities. Gender-sensitive engagement and awareness campaigns. Incentives for participation (competitions, recognition schemes).
Technological & Operational	Performance of removal devices under variable hydrological and climatic conditions may differ from lab tests.	Multi-site testing (Spain, Greece, Croatia) across different river/delta conditions. Life-cycle assessment and ecotoxicity validation ensure robustness. Contingency: deploy multiple solutions in parallel.
Environmental & Climate	Extreme weather events (floods, storms) could overwhelm removal systems or monitoring infrastructure.	Flexible and modular deployment of technologies; insurance of equipment where relevant; backup monitoring protocols (manual surveys, citizen science) for data continuity.
Legal & Regulatory	Differences in national waste, water and fisheries regulations may complicate scaling and replication across basins.	Legal analysis and policy briefs developed under WP8. Engagement with MSFD Technical Group on Marine Litter to ensure compliance and harmonisation.
Market & Replication	Slow uptake of solutions by ARs beyond pilots due to limited administrative capacity or risk aversion.	Cascade funding scheme combined with mentoring and technical assistance. Development of replication handbooks and simplified procurement templates.
Knowledge & Data Integration	Fragmentation of monitoring data and lack of interoperability with existing systems.	Mandatory use of MSFD TG ML protocols, EMODnet deposition, DTO interoperability checks. Training for authorities on data use and integration.

2.2 Measures to maximise impact - Dissemination, exploitation and communication

#@COM-DIS-VIS-CDV@#

2.2.1 First version of Plan for the Dissemination and Exploitation including Communication activities

Centralised and early planning of dissemination, exploitation, and communication at both consortium and partner level is a cornerstone of RIPPLE's impact strategy. Innovation and impact management are considered a key activity in RIPPLE and follow a multidimensional, customised, and impact-oriented approach, aiming to achieve optimal visibility through broad communication, targeted dissemination, and structured exploitation pathways,

thereby securing long-term sustainability and maximum impact. Within RIPPLE, WP7 is fully dedicated to communication, dissemination, exploitation and networking activities, synthesised in the periodically updated Plan for the Dissemination and Exploitation including Communication activities (PDEC), with a first version delivered in M6. Updates will be prepared annually to reflect progress in scientific outputs, engagement processes, and exploitation opportunities. Further sustainability, replication, and up-scaling activities are embedded in WP4 (Toolbox consolidation and uptake frameworks) and WP6 (pilot demonstrations and validation), while replication through ARs is anchored in WP7 (T7.4) to ensure basin-wide impact. RIPPLE has already identified key dissemination and communication target groups, which include European, national and regional policy makers, municipalities and river basin authorities, civil society organisations, the scientific community and academia, SMEs and innovators in environmental technologies, as well as schools, youth groups and NGOs engaged in education and awareness. These groups will be confirmed and expanded through a structured stakeholder analysis in the first version of the PDEC. Each target group will be linked to specific Key Exploitable Results (KERs) of the project, thereby aligning mutual expectations and maximising uptake. This approach will allow the consortium to design tailored messages and actions, ensuring that results are not only visible but actively translated into practice, policy and markets.

Communication activities will focus on promoting the project and its expected outcomes in a way that is clear, consistent, and accessible to diverse audiences. A Communication & Dissemination Committee (CDC), comprising the coordinator, WP7 leader, and one representative from each partner, will guide external communication, coordinate content production, and assure the quality of outreach materials. At consortium level, the CDC will define core messages, select appropriate channels, and ensure coherence across all project outputs. Distinct branding and visual identity, including templates and design guidelines, will ensure that RIPPLE maintains a professional and recognisable presence across all platforms. A Communication Toolkit will be developed for partners, including templates, factsheets, and standardised messages, while also providing adaptable formats for local communication in pilot sites and ARs. The project website will act as the main communication hub, providing up-to-date information on objectives, pilots, achievements, publications and datasets. It will be complemented by social media accounts (Twitter/X, LinkedIn, YouTube) managed according to a consortium-wide Social Media Strategy, ensuring regular visibility and interaction with key audiences. Press and news releases will be prepared at milestones, shared through partners' communication departments, and distributed to regional and national media outlets. Audio-visual material will be produced to illustrate the Toolbox, its demonstrations, and the stories of local communities; these will be released on the project website, promoted through social media, and adapted for outreach by schools and NGOs. Particular attention will be given to local communication activities in the pilot sites (Spain, Greece, Croatia) and the ARs, recognising that community ownership is essential for sustainability. Local focal points will be appointed by partners in each pilot and Associated Region to ensure that communication is delivered in local languages, adapted to cultural contexts, and embedded into community structures. Local plans will include school campaigns, participatory videos, exhibitions, and community storytelling activities. These local communication plans will be annexed to the PDEC and will empower municipalities, NGOs, and schools to act as ambassadors and multipliers of the project's vision.

Dissemination activities in RIPPLE will aim at i) sharing and promoting the results and achievements of the project while engaging with and facilitating the active participation of relevant stakeholders, and ii) creating synergies and structured networking with other EU projects and Mission Ocean initiatives. The overarching goal is to make RIPPLE results visible, accessible, and actionable for all audiences and stakeholders who may use them, including the scientific community, European, national, regional and local policy makers, civil society organisations, SMEs, and water and waste authorities. The large number of scientific, technical, and policy partners in the consortium provides a fertile ground for multi-disciplinary contributions, which will be actively promoted and encouraged. A scientific publication strategy will be developed jointly by all partners, including guidance on writing responsibilities, peer review, timing of submissions, and authorship. Open Access will be guaranteed in line with Horizon Europe rules, and a dedicated budget has been allocated to cover publication costs. Scientific results will be published in peer-reviewed journals across disciplines (marine ecology, freshwater hydrology, environmental engineering, social sciences), while all RIPPLE deliverables (policy briefs, reports, datasets, playbooks) will be made open access via the project website and stored in trusted public repositories such as Zenodo and EMODnet (as detailed in Section 1.2.7). Participation in scientific conferences will be a major dissemination channel, with partners presenting results through oral presentations, posters, and interactive sessions at national and international events (e.g. EGU, ICES, European Maritime Day, Mission Ocean Assembly, MSFD TG ML meetings). This will create opportunities for active dialogue, exchange of methodologies, and knowledge sharing with peers, strengthening the European evidence base on plastic litter. A central pillar of dissemination will

be the translation of science into policy and practice. RIPPLE will ensure that research outputs are distilled into policy briefs, adoption roadmaps, and standardisation inputs that can be directly used by municipalities, river basin authorities, and EU policy fora such as the MSFD Technical Group on Marine Litter. This science-to-policy translation is essential for embedding Toolbox solutions into governance, thereby maximising their long-term uptake. Specific learning materials – such as modular training packages for municipalities and guidance notes for port and fisheries authorities – will be prepared and made available through the Toolbox platform. External stakeholders will be systematically engaged to ensure that the knowledge generated and lessons learned within the project extend beyond the consortium and pilot regions. To this end, RIPPLE will develop an open learning environment, integrated into the Toolbox platform, serving as a one-stop shop that provides access to all project materials: harmonised monitoring protocols, citizen science guidelines, prevention playbooks, and training modules. This online environment will be promoted through webinars, video teasers, and social media campaigns, and linked to European platforms such as the Mission Implementation Platform and the European School Education Platform, ensuring visibility across broader audiences. RIPPLE will also engage external actors through policy masterclasses and stakeholder webinars. These will target public officials, NGOs, and industry actors, enabling them to test the Toolbox resources and adapt them to their contexts. ARs, selected through an open call, will be central actors in this dissemination process, not only receiving mentoring and training but also serving as multipliers who showcase adaptation and replication of solutions. Finally, dissemination will be reinforced through networking and cooperation with other projects. RIPPLE will actively link with Mission Ocean initiatives such as Plastic Pirates and WATER4ALL, and with related Horizon and Interreg projects (e.g. INSPIRE, REMEDIES, CLEANATLANTIC). Joint workshops, thematic webinars, and cross-project dissemination campaigns will ensure synergies, avoid duplication, and maximise the visibility of results at European level. To anchor dissemination in long-term structures, RIPPLE will establish a RIPPLE Forum, modelled as an open space for knowledge exchange, co-creation, and dialogue. The Forum will host webinars, thematic meetings, and Q&A sessions with external stakeholders, enriching project deliverables and contributing to policy recommendations. RIPPLE will ensure that the results are not only disseminated but internalised within policy, practice, and community networks across the Mediterranean basin and beyond.

Activity	Description	KPI
Project Website	Central hub presenting project objectives, pilots, partners, and outputs (publications, datasets, policy briefs, events, audio-visual material). Regularly updated with news and results.	≥ 500 visits per month by Year 2; ≥ 750 by project end
Dedicated social media accounts	Project-level Twitter/X, LinkedIn, and YouTube channels, complemented by partners' existing accounts. Used to promote deliverables, publications, events, and success stories from pilots and ARs.	≥ 1,000 followers across channels by project end
Audio-visual material	1 professional video explaining key Toolbox features and results; 1 storytelling video collecting local audio-visual material from pilot sites and ARs.	≥ 5,000 combined views
Press releases	Consortium-wide press releases at milestones (kick-off, ARs scheme launch, pilot demonstrations, final conference), distributed through partners' press offices.	≥ 4 press releases; first at kick-off
Local communication plans and activities	Locally tailored communication in pilots and ARs, e.g. school exhibitions, participatory video production, community storytelling, newsletters, blogs, podcasts.	≥ 1 local activity per pilot/Associated Region per year (≈ 9–12 total)
Participation in international and local forums/conferences	Presentation of results at scientific and policy events (e.g. European Maritime Day, MSFD TG ML, EGU, Mission Ocean Assembly, Interreg MED events). Organisation of dedicated sessions at major EU conferences.	≥ 8 conferences/public events; ≥ 1 dedicated session organised
Publications in scientific journals	Peer-reviewed articles across marine ecology, freshwater hydrology, environmental engineering, circular economy, and governance. All papers open access.	≥ 6 publications

Joint activities with relevant EU and projects initiatives	Cooperation with Mission Ocean initiatives (Plastic Pirates, WATER4ALL) and Horizon/Interreg projects (INSPIRE, REMEDIES, CLEANATLANTIC). Joint workshops and knowledge exchange actions.	≥ 4 projects engaged
RIPPLE Forum actions	Thematic webinars, workshops, Q&A sessions with external stakeholders to foster engagement, share knowledge, and co-develop policy recommendations.	≥ 4 actions
Final conference in Brussels part of the New European Bauhaus Festival	High-profile closing event presenting project outcomes, Toolbox, and replication strategies. ARs and external stakeholders invited.	≥ 120 participants

Exploitation strategy A stepwise approach to exploitation management will ensure that the Key Exploitable Results (KERs) of RIPPLE are fully aligned with the project's expected impacts, systematically documented, assessed, protected, and pursued by individual partners and at consortium level. Exploitation will encompass products and services (e.g. validated monitoring and removal technologies), research and development (harmonised datasets, circular entrepreneurship pilots), and policy making (policy briefs, adoption roadmaps). Established innovation management tools and workshops will be used to structure exploitation pathways, with clear monitoring of uptake during the project and in the four years following its completion. The approach will be implemented in two main phases: **Phase 1: Capacity Building and Strategy Validation** At project inception, a capacity building session will outline rules for IP management under Horizon Europe and introduce collaborative IP management strategies. An IP and Exploitation Manual will provide guidance on challenges, risks, and opportunities, and will detail rules agreed in the Consortium Agreement. A first Exploitation Strategy Workshop will validate the list of KERs and their associated exploitation pathways. Exploitation strategies will be reviewed annually and refined based on stakeholder feedback and pilot validation. **Phase 2: Exploitation and Sustainability Planning** Exploitation pathways will be further developed at consortium level, taking into account individual partner perspectives identified through structured Innovation Questionnaires. A needs–contributions–benefits–risks matrix will map interactions, contributions, and dependencies across partners, strengthening ownership and long-term commitments. A Results Ownership List (ROL) will be created to define individual and joint ownership of KERs. Successive Exploitation and Sustainability Planning Workshops will develop action plans for each KER, documented in a consolidated Exploitation Roadmap. This Roadmap will include time planning, ownership details, IP status, value propositions, market segments, risks, competitors, and legal considerations. It will also define concrete sustainability measures, including integration of the Toolbox into public procurement practices, links with financing instruments and community-driven replication through ARs. At local level, RIPPLE will work with pilot sites and ARs to prepare sustainability and exploitation plans. These will include long-term visions for uptake by municipalities and regional authorities, building on the Toolbox as a practical governance instrument. ARs will be supported to co-develop replication roadmaps, enabling them to embed Toolbox solutions into their policies and financing pipelines, while also providing third parties with models for replication beyond the consortium.

KER	Description	Exploit. Pathway	IPP	Target Groups	KPI
KER1: Mediterranean Toolbox	Integrated set of monitoring, prevention, removal and circular economy solutions.	Publicly available Toolbox maintained by consortium partners; used by municipalities and NGOs; promoted via EU platforms.	Copyright ; open licence (CC BY 4.0).	Public authorities, municipalities, NGOs, ports, water agencies.	≥ 200 Toolbox users by project end; ≥ 3 confirmed commitments for uptake post-project
KER2: Harmonised Monitoring Protocols & Datasets	Protocols for monitoring plastics from river to sea; FAIR datasets.	Published datasets uploaded to EMODnet/DTO; used by national monitoring schemes.	Copyright ; open data licence.	Scientific community, monitoring agencies, ministries.	≥ 20 datasets uploaded; ≥ 3 protocols adopted by agencies
KER3: Validated Removal Technologies & O&M Playbooks	Tested removal devices with operational manuals.	Uptake by municipalities via procurement; guidance for SMEs.	Copyright ; potential patents.	Municipalities, ports, SMEs, waste operators.	≥ 2 technologies validated; ≥ 3 O&M

					playbooks published
KER4: Prevention Playbooks & Procurement Templates	Guidance for public procurement and community-based prevention.	Templates integrated into municipal procurement practices; shared with NGOs and schools.	Copyright ; Creative Commons licence.	Local/regional authorities, NGOs, schools.	≥ 3 prevention playbooks delivered; ≥ 3 procurement templates tested
KER5: Education & Training Modules	Online training packages for municipalities, schools, NGOs.	Hosted in Toolbox; integrated into EU platforms (EPALE, School Education Gateway).	Copyright ; Creative Commons licence.	Schools, NGOs, municipalities, training centres.	≥ 3 training modules published; ≥ 200 learners engaged
KER6: Circular Entrepreneurship Pilots & Models	Models for SMEs/start-ups valorising plastics.	Promoted through regional innovation agencies; business cases shared via Toolbox.	Copyright ; potential trademarks.	SMEs, start-ups, innovation agencies.	≥ 2 business models documented; ≥ 2 SMEs supported
KER7: Replication Roadmaps for ARs	Tailored roadmaps for ≥3 ARs.	Co-developed with ARs; shared as guidance for other territories.	Copyright ; open licence.	Municipalities, regional authorities, NGOs.	≥ 3 replication roadmaps completed; ≥ 6 peer-to-peer exchanges held
KER8: Policy Briefs & Adoption Roadmaps	Evidence-based recommendations linked to EU Zero Pollution, MSFD, CEAP.	≥ 6 policy briefs delivered; presented to EU/national authorities; uptake roadmaps	Copyright .	EU institutions, ministries, municipalities, NGOs.	≥ 6 policy briefs; ≥ 3 policy uptake commitments secured

2.2.2 Strategy for IP management

RIPPLE will generate a range of valuable intellectual property, including harmonised monitoring protocols and datasets, validated removal technologies, prevention playbooks, education and training modules, and replication roadmaps. To maximise the impact of these results and ensure their responsible and equitable use, the consortium will establish a comprehensive IP management strategy that covers the period before, during, and after the project. This strategy will ensure that ownership is clearly defined, that protection measures are applied where appropriate, and that exploitation is supported by transparent and fair rules. Before the project start, the Consortium Agreement will set out the provisions for the use of Background IP needed for implementation. It will also clarify access rights and establish a clear framework for how such Background IP can be used in the future, in particular in relation to exploitation. This will provide legal certainty from the outset and create a shared basis for collaboration. During the project, IP opportunities will be systematically identified, assessed and captured. Potential protection measures may include patents for technology components such as removal devices, design rights for innovative monitoring solutions, copyright for prevention playbooks, training materials and replication roadmaps, as well as the use of trade secrets where commercially sensitive know-how is involved. Ownership of results will be clarified as early as possible, and where results are developed jointly, dedicated Joint Ownership Agreements will be concluded. Freedom-to-operate analyses and IPR checks will be performed when appropriate, ensuring that technologies and tools validated in RIPPLE can be transferred to the market without legal or competitive obstacles. Partners will also use this stage to identify potential exploitation partners and investors, preparing results for their uptake by municipalities, SMEs, NGOs and policy actors. After the project, the consortium will implement post-project agreements, including licensing agreements, transfer of IP to specific partners or stakeholders, and collaboration arrangements to further develop or scale solutions. Management of IP will include a transparent approach to costs and revenues, ensuring fair allocation and benefit-sharing across partners. Exploitation will include commercialisation of removal technologies by SMEs, further research and innovation based on open datasets, integration of prevention playbooks and procurement templates into municipal practices, and uptake of policy briefs and roadmaps by national and EU institutions. To support this process, RIPPLE will prepare an IP and Exploitation Manual and deliver capacity building sessions for all partners. These will provide practical guidance

on managing access rights, ownership, protection and use of results both during and after the project. The Manual will also include provisions for managing community-based contributions, such as citizen science data and Associated Region roadmaps, ensuring that these contributions are appropriately acknowledged and that rights are clearly attributed. The provisions set out in the Consortium Agreement will thus provide a flexible and efficient framework for the ownership, protection and use of intellectual property, both existing and generated during the project. Joint Ownership Agreements and post-project provisions will clarify responsibilities for the management of IP, including who will bear the costs of protection, how exploitation will be pursued, and how revenues will be shared. Finally, in line with Horizon Europe requirements, the consortium will establish and maintain a Results Ownership List, which will be included in the final periodic report, clearly identifying the owner or co-owners of each Key Exploitable Result. This will enable both the consortium and individual partners to pursue market opportunities, policy uptake and community replication of RIPPLE results in a transparent and sustainable manner. In addition to internal consortium management, RIPPLE will establish clear provisions for results generated with the involvement of ARs and citizen science contributors. ARs, supported through cascade funding, will co-develop replication roadmaps and adapt Toolbox elements to their local contexts. While these regions will retain full rights to their locally generated data and materials, the consortium will secure the right to integrate and disseminate these results as part of the Toolbox to ensure European-wide impact. Citizen science contributions, such as data collected through schools, NGOs, and community groups, will be openly licensed under Creative Commons agreements to guarantee transparency, accessibility, and recognition of contributors. The intention is that UZAF, ICCS and CSIC, together with pilot municipalities, act as anchors for maintaining the Toolbox and dashboards, while TEAMSLAB, DUNEA and KYVELI facilitate continuation of the AR network. These responsibilities will be detailed and confirmed during the project, ensuring that long-term impact is embedded in local governance and policy frameworks. This approach balances the need for broad accessibility with respect for local ownership, ensuring that the Toolbox remains both scalable and inclusive while preserving fair attribution of all actors involved.

IP Measure	Partner(s)	Expected Protection Type	Link to Exploitation
Background IP definition & access rights	Coordinator + all partners	Defined in Consortium Agreement	Ensures legal certainty and partner access for implementation and future use
Identification of new IP (KERs)	WP leaders (WP2–WP6)	Case-by-case: patents (tech), copyright (playbooks, datasets), design rights (tools), trade secrets (know-how)	Prepares results for uptake by municipalities, SMEs, NGOs, and policy actors
Ownership & Joint Ownership Agreements	Coordinator + Innovation/Exploitation Manager	Legal agreements under Horizon Europe	Clarifies responsibilities and rights, enabling smooth exploitation
IP & Exploitation Manual (D7.2)	WP7 leader	Internal guidance document	Provides partners with common rules for protection, access and use of results
Capacity building sessions on IPR	WP7 leader + external experts	Training, no formal protection	Strengthens partners' ability to protect and exploit results during and after project
Freedom-to-operate and IPR checks	SMEs/tech providers + WP3 leader	Patents, trade secrets, copyright	Ensures validated removal technologies can enter market without conflicts
Results Ownership List (ROL)	Coordinator + all partners	Finalised in last periodic report	Identifies owners/co-owners of each KER, securing clarity for post exploitation
Post-project agreements	Coordinator + partners holding IPR	Licensing agreements, collaboration contracts, transfer agreements	Enables further research, commercialisation, and policy uptake
Citizen science and ARs contributions	WP7 leader + ARs	Creative Commons licences, copyright	Guarantees open access, respecting local ownership and recognition

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2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Overcome the lack of integrated tools for public authorities to address plastics from river to ocean. Harmonise monitoring protocols and data flows across Mediterranean basins. Validate affordable prevention and removal solutions adapted to local authorities. Reduce fragmentation between science, policy, and practice in tackling marine litter. Translate scientific evidence into actionable policy and procurement pathways	R1: Mediterranean Toolbox integrating monitoring, prevention, removal, and circular solutions. R2: Harmonised monitoring protocols and FAIR datasets uploaded to EMODnet/DTO. R3: Validated removal technologies with operational playbooks. R4: Prevention playbooks and procurement templates for municipalities. R5: Education and training modules for schools, NGOs, and authorities. R6: Circular entrepreneurship pilots and business models. R7: Replication roadmaps co-developed with ARs. R8: Policy briefs and adoption roadmaps aligned with EU strategies.	Dissemination: Open Access publications; datasets uploaded to EMODnet/DTO; presentations at international conferences; policy briefs distributed to EU/national authorities. Networking: Joint activities with Mission Ocean initiatives (Plastic Pirates, WATER4ALL, Interreg projects). Communication: Project website, social media, press releases, professional video and storytelling clips from pilots and ARs; local communication activities in schools and municipalities. Exploitation: Exploitation Roadmap, IP & Exploitation Manual, Results Ownership List; capacity building and workshops to secure uptake of Toolbox solutions, policy briefs, and replication roadmaps.
TARGET GROUPS	OUTCOMES	IMPACTS
Municipalities and river basin authorities in pilot sites and ARs Ports and fisheries authorities Regional and national ministries responsible for environment, water, waste, and marine policy, aligning with MSFD and Zero Pollution targets. SMEs, FabLabs, and start-ups developing and piloting circular reuse of collected plastics. Schools, NGOs, and citizen groups. Scientific and monitoring organisations applying harmonised TG ML protocols and sharing open datasets. Mission Ocean platforms and EU initiatives and projects	Public authorities access a co-designed Mediterranean Toolbox with harmonised monitoring, validated removal technologies, and engagement measures. Pilots in Spain, Greece, and Croatia demonstrate reduction in plastic fluxes through combined monitoring, removal, and prevention interventions. Operational playbooks, maintenance models, and cost-effectiveness assessments accelerate uptake of innovative solutions by municipalities and SMEs. Harmonised datasets (micro/macro plastics) generated with TG ML protocols. Replication with ≥ 3 ARs contributes directly to Mission Ocean and Zero Pollution targets Dashboards and APIs allow authorities to visualise and integrate data into planning; synergies with EU initiatives ensure long-term knowledge sharing.	Demonstrate and replicate a Toolbox in 3 pilots and ARs, reducing plastic fluxes. Strengthen governance by aligning monitoring, prevention, and removal with EU Zero Pollution goals. Engage citizens, schools, and NGOs to drive awareness and behavioural change. Validate cost-effective removal technologies and digital dashboards for local uptake. Generate new scientific knowledge with harmonised protocols and open datasets. Stimulate circular economy models with SMEs and fisheries reusing plastics. Deliver policy briefs and adoption roadmaps supporting Mission Ocean targets

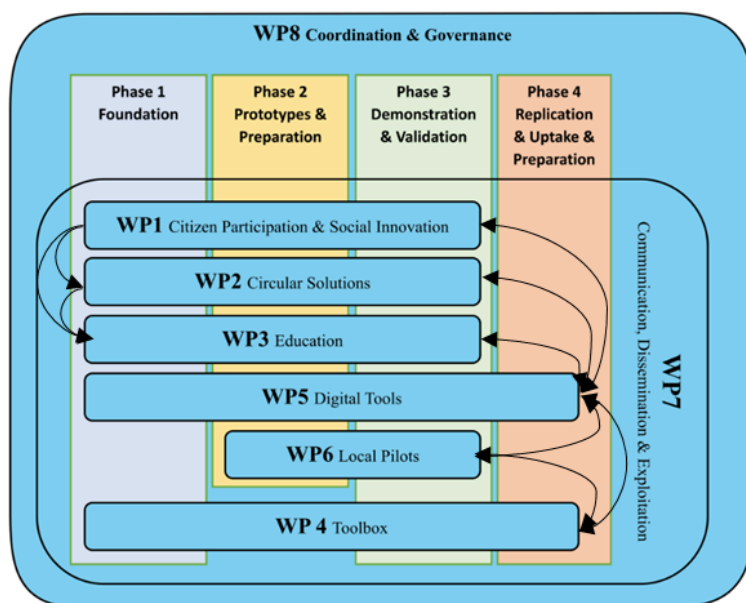
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3. Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

The RIPPLE work plan is composed of eight interlinked Work Packages (WPs), structured around the four methodological phases of the project. This design ensures scientific robustness, operational feasibility, and basin-level scalability, with feedback loops between pilots, digital platforms, and stakeholder processes guaranteeing continuous refinement and alignment with Mission Ocean objectives. In **Phase 1 (Foundation, M1–12)**, WP1 (Citizen Participation & Social Innovation) mobilises municipalities, schools, NGOs, and communities to co-design prevention measures through stakeholder mapping, citizen science protocols, and participatory workshops. In parallel, WP2 (Circular Solutions) maps and assesses existing open-source tools, while WP3 (Education) develops education frameworks and training modules, drawing on established methodologies such as Universal Design for Learning and whole-school approaches. WP5 (Digital Tools) begins developing harmonised monitoring protocols aligned with MSFD TG ML and prepares data pipelines to ensure interoperability with EMODnet and DTO. WP4 (Toolbox) initiates the design of its architecture to integrate these outputs. Together, these activities provide the conceptual and operational foundation for the entire project. In **Phase 2 (Prototyping & Preparation, M6–18)**, WP2 progresses to co-design sessions in FabLabs and makerspaces, producing prototypes tested with entrepreneurs and civic groups. WP3 pilots initial education modules, while WP5 develops and tests dashboards and interfaces with municipalities to ensure usability. WP6 (Local Pilots) begins preparatory work in Spain (Besòs River, urban–industrial pressures), Greece (Remathia Stream, sub-urban with strong citizen-science traditions), and Croatia (Neretva River, cross-border and biodiversity-rich). In each pilot, governance arrangements, monitoring infrastructure, and citizen engagement processes are established, ensuring readiness for demonstration. WP1 supports early buy-in by rolling out citizen science and awareness campaigns in schools and communities. In **Phase 3 (Demonstration & Validation, M12–30)**, WP6 leads full-scale deployment of removal technologies (such as Floating Booms, Trash Traps, Bio barriers,) under real conditions. These are validated through LCA and ecotoxicity analyses coordinated with WP4. In parallel, prevention measures are

implemented: fishing-for-litter and municipal fixes in Besòs, school campaigns and waste-system improvements in Remathia, and port reception upgrades in Neretva. WP2 supports circular transformation pilots that convert collected plastics into usable products, linking environmental action to social innovation and entrepreneurship. WP5 ensures continuous monitoring, with datasets openly published to EMODnet within three months, while WP4 consolidates validated results into playbooks, procurement templates, and replication frameworks. WP3 evaluates and iterates the educational programmes to ensure long-term impact. By the end of this phase, the Toolbox is populated with operational, authority-ready content. In **Phase 4 (Replication & Uptake, M18–36)**, WP7 (Communication, Dissemination & Exploitation) launches the ARs scheme, providing FSTP of up to €100,000 per region alongside structured mentoring



and technical assistance. Replication roadmaps and uptake strategies are co-developed with at least three ARs, extending results beyond the pilots. WP7 also delivers adoption roadmaps, policy briefs, and financing pathways to secure sustainability, while WP4 and WP5 finalise the Toolbox as a digital platform fully interoperable with EMODnet and DTO. The **linkages between WPs are deliberate and reinforcing**: WP1–3 engage citizens, map solutions, and build capacity; WP2–5 advance technical and digital tools; WP6 validates them in pilots; WP4 integrates all outputs into the Toolbox; WP7 drives replication and policy uptake; and WP8 ensures consistent coordination. This structure ensures that each WP both contributes to and benefits from others, creating a coherent framework that moves from local co-creation to basin-wide scaling.

		YEAR 1												YEAR 2												YEAR 3												YEAR 4											
	M	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
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3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

The RIPPLE consortium includes **16 full partners and 2 associated partners** from 8 countries (Croatia, Spain, Greece, Bosnia & Herzegovina, etc.). It combines **universities and research institutes, SMEs and makerspaces, regional development agencies, and NGOs**, ensuring a full innovation cycle: from citizen engagement and co-design → to technical validation in pilots → to replication via Associated Regions. The consortium is structured in two main groups: **1. RIPPLE Technical & Innovation Partners (TIPs)**: Universities, research centres, SMEs, FabLabs and digital developers that bring scientific, technical, and entrepreneurial expertise (UZAF, UNIDU, CSIC, ICCS, FLZG, TransfoLAB BCN, Decode, TechnAI, Institute of Entrepreneurial Education & Innovation, University of Mostar – assoc.); **2. RIPPLE Territorial & Engagement Partners (TEPs)**: Local/regional authorities, NGOs, and creative hubs ensuring citizen participation, governance relevance, and pilot implementation (DUNEA, KYVELI, Commonsplace, TeamLabs,, Lichen, CoLab House, ZTKDU, Agency for Watershed of Adriatic Sea – assoc.).

Technical & Innovation Partners (TIPs)

UZAF (UZAF, Croatia): Coordinator, leader of WP4 (Toolbox) and WP8 (Management). Expertise in replication strategies, governance, EU project coordination. **University of Dubrovnik (UNIDU, Croatia)**: Marine and coastal sciences; supports pilot preparation and education evaluation. **CSIC (Spain)**: EU reference for environmental monitoring; leads MSFD TG ML compliant monitoring (WP5). **ICCS (Greece)**: Leader of WP5 (Digital Tools & Platform); expertise in DTO interoperability and decision support. **FabLab Zagreb (Croatia)**: Maker and prototyping expertise; leads T2.3 and WP7 communication; connects open-source tools with citizen-driven pilots. **TransfoLAB BCN (Spain)**: WP2 leader; creative ecosystem in Barcelona specialised in circular economy, design, reuse, and fabrication; brings extensive expertise in material innovation and circular entrepreneurship. **Decode IKE (Greece)**: Supports makerspaces, digital fabrication, and community pilots. **TechnAI (Spain)**: AI-driven dashboards, dissemination, and exploitation tools.. **Institute of Entrepreneurial Education & Innovation -IEEI (Croatia)**: Training and entrepreneurship integration for circular pilots. **University of Mostar (Bosnia & Herzegovina, assoc.)**: Academic expertise in the Neretva basin, supporting monitoring and training.

Territorial & Engagement Partners (TEPs)

DUNEA (Croatia): Regional Development Agency; ensures governance, replication, and uptake in the Croatian pilot. **KYVELI (Greece)**: Local authority; responsible for pilot implementation in Greece. **TeamLabs (Spain)**: Mandated by the Barcelona City Council, expertise in circular economy and entrepreneurship; contributes to pilot preparation, replication, and training; **Commonspace (Greece)**: WP6 leader; participatory governance and pilot management. **Lichen (Spain)**: WP1 leader; social innovation lab focused on citizen empowerment and participatory processes. **CoLab House (Greece)**: WP3 leader; creative hub dedicated to circular economy, social innovation and sustainability education. **ZTKDU (Croatia)**: Community of Technical Culture; mobilises schools, youth, and communities for pilots and monitoring. **Agency for Watershed of the Adriatic Sea (Bosnia & Herzegovina, assoc.)**: Ensures policy alignment and governance capacity for the Neretva basin pilot.

What makes RIPPLE unique is its ability to merge grassroots creativity with scientific rigour: from citizen-driven makerspaces and FabLabs prototyping circular solutions, to universities and research institutes validating them under EU standards, and regional authorities guaranteeing real uptake. Few consortia can bridge this full cycle — from imagination to institutionalisation — in three countries and replicate it across the Mediterranean. This capacity to connect bottom-up innovation with top-down governance is our strength, and the reason we can deliver systemic and lasting impact. A further strategic advantage is the inclusion of two partners from Bosnia and Herzegovina (University of Mostar and the Adriatic Watershed Agency), who ensure that impacts in the Neretva basin — a river system rooted in Herzegovina but flowing into the Adriatic — are addressed at the source. This upstream–downstream integration guarantees that the project’s impact is not only visible at the coastline, but embedded in the governance and communities where the problem begins.

Scientific backbone (UZAF, UNIDU, CSIC, University of Mostar).

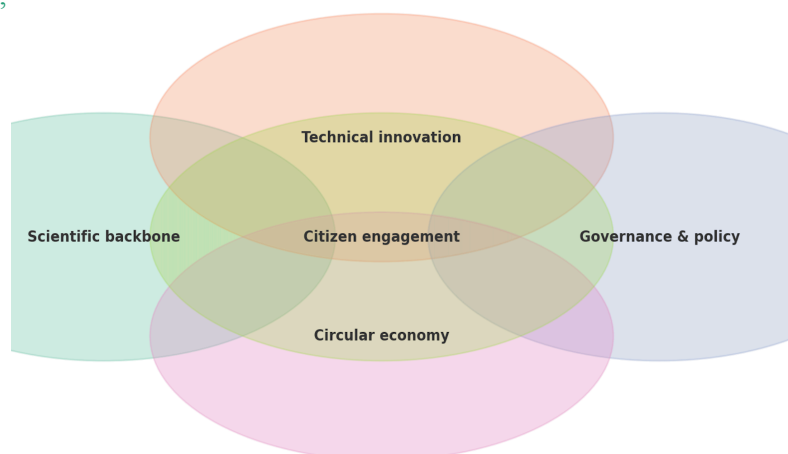
Technical innovation (FLZG, TransfoLAB BCN, ICCS, Decode, TechnAI).

Local governance and policy (DUNEA, KYVELI, Watershed Agency, TeamLabs).

Circular economy and entrepreneurship (TransfoLAB BCN, TeamLabs, IEET).

Citizen engagement and education (Lichen, CoLab House, Commonsplace, ZTKDU).

RIPPLE Consortium Complementarities



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Tables for section 3.1

Table 3.1a: List of work packages

WP No	Work Package Title	LP No	LP Short Name	PM	Start Month	End month
WP1	Citizen Participation & Social Innovation	11	Lichen	90	1	36
WP2	Co-Design of Circular Solutions and Open-Source Tools	8	TransfoLAB BCN	94	1	30
WP3	Development of Sustainability Education Programs	2	CoLab House	102	1	48
WP4	Toolbox and Replication Strategies Development	1	UZAF	121	1	48
WP5	Digital Tools and Platforms for monitoring and open data sharing	6	ICCS	121	6	48
WP6	Local Pilot Programs	3	Commonspace	162	12	48
WP7	Communication, Dissemination, and Exploitation	10	FabLab ZG	113	1	48
WP8	Coordination and Governance	1	UZAF	61	1	48

Table 3.1b: Work package description

Work package number	1 - Lead Partner: Lichen
Work package title	Citizen Participation & Social Innovation
WP1 specific objectives are: 1) To co-design communities-centered methodologies, tools and approaches for citizen science and solutions development processes within the project. 2) To engage local ecosystems to identify challenges and co-design integrated solutions with municipalities, experts, companies and citizens. 3) To design a participatory approach within citizen and school programmes, awareness campaigns, citizen	

science and co-creations sessions. 4) To empower communities and municipalities with education, participation, circular solutions and social innovation.

Description of Work: Task 1.1: Stakeholders participation *Lead: Commonsplace, Participants: CoLab House, Lichen, TransfoLAB BCN, IEEI, UNIDU, DUNEA; Months: 1-12* This task will design and implement participatory processes with municipalities, companies, schools, NGOs, and local communities to identify motivations, challenges, risks, and opportunities related with plastic pollution in their local contexts. Activities will include stakeholder mapping, participatory diagnosis and collective workshops in the three pilot sites (Besòs, Remathia, Neretva). Outputs will be consolidated in a “Participatory Diagnosis Report” on each pilot site, highlighting dreams, fears and expectations. These reports will guide the design of co-created solutions and ensure alignment with municipal waste strategies and citizen priorities. D1.1

Task 1.2: Stakeholder Collaboration Framework *Lead: Lichen; Participants: ICCS, Commonsplace, TechnAI, IEEI, ZTKDU; Months: 3-12* This task will develop a project framework for cross-cutting collaboration, connecting municipalities, schools, companies, NGOs, Communities and project partners. The framework will define participation objectives, expectations, principles, decision-making methods, and governance, including gender equality, inclusion of vulnerable groups, and RRI principles. Tools such as online collaboration platforms, digital dashboards, and participatory design methods will be embedded. The framework will serve as a reference for pilots and Associated Regions. D2.1

Task 1.3: Citizen Engagement Strategies *Lead: Lichen; Participants: All pilot partners; Months: 6-36*

Citizen engagement strategies will be co-designed and implemented across project phases. These will include citizen science protocols, educational resources, behavioural nudges, and awareness campaigns linked to local events and festivals. Each pilot site will test tailored approaches to reflect its socio-cultural context. The strategies will reach a wide audience through schools, community groups, NGOs, and municipal events, with the ambition to directly engage more than 500 citizens and students across the three pilots. Outputs will include citizen science protocols aligned with MSFD TG ML guidelines, a comprehensive package of educational resources, and at least six awareness campaigns designed to shift behaviours and attitudes. Success will be tracked through defined project KPIs, ensuring evidence-based validation of the engagement approaches D1.3

Work package number

2 - Lead Partner: TransfoLAB BCN

Work package title

Co-Design of Circular Solutions and Open-Source Tools

WP2 specific objectives are: 1) to Co-design prevention and collection measures with authorities, industry, and citizens, 2) Engage local ecosystems to identify risks and co-design integrated solutions, 3) Run citizen and school programmes, awareness campaigns, and behavioural nudges, 4) Co-develop procurement templates and waste-system improvements with municipalities, 5) Empower municipalities and communities with education and circular solutions, 6) Integrate citizen science and co-creation methodologies 7) Enable social entrepreneurship for circular transformation of collected plastic 8) Demonstrate sustainable and safe removal technologies.

Description of Work: Task 2.1: Mapping existing open-source low-tech and low-cost solutions *Lead: CoLab House; Participants: TransfoLAB BCN; Decode Fab Lab, FLZG, TeamsLab; Months: 1-12* Secondary research to identify existing open-source low-tech and low-cost solutions that have been already tested and verified. This task will identify, screen, and evaluate existing open-source low-tech and low-cost solutions relevant to plastic prevention, collection, and transformation. CoLab House will coordinate a desk review of initiatives such as Floating Booms, Trash Traps, Bio Barriers, Precious Plastics, and mobile FabLabs. Decode Fab Lab and FLZG will contribute expertise in prototyping and fabrication, while TransfoLAB BCN and TeamLabs will assess circularity and community relevance. A framework will be developed to evaluate applicability by demographic, geographic, and environmental conditions of the pilot sites. Citizen-based solutions (Repair Cafés, Material Libraries, Pop-up Collection Points) will be mapped and analysed for integration potential. Outputs will provide a prioritised list of candidate solutions and criteria for piloting. At least 30 solutions screened (10 per category: monitoring, removal, circular transformation), 1 evaluation framework developed and applied to all, 3 prioritisation reports prepared (one per pilot context). D2.1

Task 2.2: Co-design sessions in makerspaces/fab-labs *Lead: TransfoLAB BCN; Participants: CoLab House, Commonsplace, FLZG, ZTKDU, Teamlabs; Months: 6-18* - this task will implement pilot-scale circular transformation activities involving citizens, social entrepreneurs, and SMEs. IEEI will coordinate

pilots where recovered plastics are transformed into community products (e.g. furniture, educational tools, urban infrastructure elements). TransfoLAB BCN and ZTKDU will provide maker expertise, while Lichen ensures participatory engagement and social innovation. CoLab House and Commonsplace will connect to schools, NGOs, and municipalities to broaden participation. TeamLabs Till ensure entrepreneurship pathways for viable solutions. Outputs include at least three tested community transformation pilots per site, creating a sandbox for testing technical and social feasibility. 6 co-design events across pilots (2 per country). Each event engaging 30–50 participants (makers, citizens, SMEs, schools). Output: ≥15 co-created concepts/prototypes documented. D2.2

T2.3 Prototyping & pilot testing of open-source tools *Lead: FLZG; Participants: TransfoLAB BCN, CoLab House, Decode Fab Lab, ZTKDU, Teamlabs; Months: 6-18* This task will advance selected concepts from T2.1 and T2.2 into functional prototypes. FLZG will lead iterative prototyping, supported by Decode and TransfoLAB BCN on materials and fabrication methods. Prototypes will focus on: (1) plastic collection devices adapted to pilot river-sea conditions, and (2) plastic transformation tools enabling civic and entrepreneurial use of recovered waste. CoLab House and TeamLabs will ensure co-design with end-users (municipalities, schools, SMEs). Testing will occur in controlled pilot settings with monitoring of technical performance, safety, and usability. Lessons will inform scale-up in WP6. 5–6 prototypes built and tested (2 per pilot minimum). Testing periods of 3–6 months per prototype. Safety and durability testing under local environmental conditions (river/sea interface). D2.2

T2.4 Circular transformation pilots with entrepreneurs & civic groups *Lead: IEEI; Participants: CoLab House, Commonsplace, Lichen, TransfoLAB BCN, ZTKDU, Teamlabs; Months: 12 - 24* - This task will implement pilot-scale circular transformation activities involving citizens, social entrepreneurs, and SMEs. IEEI will coordinate pilots where recovered plastics are transformed into community products (e.g. furniture, educational tools, urban infrastructure elements). TransfoLAB BCN and ZTKDU will provide maker expertise, while Lichen ensures participatory engagement and social innovation. CoLab House and Commonsplace will connect to schools, NGOs, and municipalities to broaden participation. TeamLabs will ensure entrepreneurship pathways for viable solutions. Outputs include at least three tested community transformation pilots per site, creating a sandbox for testing technical and social feasibility. 3 pilot initiatives per country (e.g., Precious Plastic hub, Repair Café, maker training), at least 10 SMEs/entrepreneurs engaged across all sites, 100+ community members involved in testing/using products. T2.3. D2.2

T2.5 Integration with validated removal technologies and procurement templates *Lead: TransfoLAB BCN; Participants: CoLab House, KYVELI MAE, TeamLabs, DUNE, ZTKDU, FLZG; Months: 12-30* - This task ensures that developed prototypes are integrated with validated removal technologies (from WP6) and linked to municipal waste-management systems. TransfoLAB BCN will lead integration, with DUNE and KYVELI ensuring governance and municipal adoption. CoLab House will co-develop procurement templates, while TeamLabs adapts them for SMEs and entrepreneurs. ZTKDU and FLZG will contribute technical validation of durability and safety. Outputs will include operational guidelines, procurement templates, and models for long-term adoption by public authorities. At least 3 validated removal technologies linked with prototypes (one per pilot), 3 procurement templates drafted and tested with municipalities, 1 operational guideline package per pilot site (for municipal adoption). Linked to T2.1. D2.1

Work package number	3 - Lead Partner: CoLab House
Work package title	Development of Sustainability Education Programs

WP3 specific objectives are: 1) Empower municipalities and communities with education and circular solutions 2) Develop sustainability education programmes for municipalities, schools and citizens, T.O.2.2 – Integrate citizen science and co-creation methodologies 3) Enable social entrepreneurship for circular transformation of collected plastic

Description of Work : Development of education and training programmes for municipalities, schools & communities. Task 3.1: Education framework design - Lead: CoLab House; Participants: TransfoLAB BCN, Decode Fab Lab , IEEI; Months: 1-12 Identify the needs and training areas of the target audiences, based on which the educational framework will be developed. Needs will be mapped through surveys and workshops in each pilot, synthesised by CoLab House. TransfoLAB will contribute sustainability and circular economy expertise, while IEEI ensures alignment with entrepreneurship education. The framework will be validated by local schools and municipalities before finalisation. We will consider existing frameworks where applicable, such as Universal Design for Learning (UDL), Whole School Approach,

Experiential Learning, Active Learning etc to remove barriers and increase the success of the educational program. D3.1

T3.2 Development of training modules - *Lead: Decode Fab Lab; Participants: CoLab House, TransfoLAB BCN, IDAEA, ICCS, Teamlabs; Months: 6-18* Based on the needs of the target audiences and the educational framework developed in T3.1 we will develop tailored education and training programmes that equip municipalities, schools, and communities with the knowledge and tools to implement sustainable practices effectively. Connecting to the outputs in WP1 and WP2 (needs, technologies, frameworks, templates), as well as synching with T5.1 and T5.2, during this task all the necessary training modules, lectures, guidebook, tutorials and practical exercises will be developed. At least 6 training modules will be developed, covering monitoring, prevention, removal, and circular transformation. Modules will include teacher kits for schools, operational handbooks for municipalities, and entrepreneurship guides for SMEs. IDAEA will validate scientific accuracy, ICCS will ensure digital integration with the platform. Materials will adapt to experiential learning and integrated in the Digital dashboard for usability and accessibility (T5.1 and T5.2). D3.1

T3.3 Evaluation & iteration of education programmes - *Lead: TransfoLAB BCN; Participants: CoLab House, Decode Fab Lab, IEEL, UNIDU, UZAF, Teamlabs; Months: 18-48*; The educational programme will undergo continuous evaluation and iteration to ensure accessibility, inclusivity, affordability, and local relevance. Evaluation will be conducted through participatory feedback loops with municipalities, citizens, and other stakeholders engaged in the three pilot use cases (Spain, Greece, Croatia). Evaluation will be based on pre/post training surveys, focus groups, and pilot testing in the three sites. TransfoLAB will lead evaluation design, UNIDU and UZAF will ensure academic rigour, while TeamLabs will refine learning methods. Iterations will be carried out annually, with updated versions of modules uploaded to the digital dashboard. D3.2

Work package number

4 - Lead Partner: UZAF

Work package title

Toolbox and Replication Strategies Development

WP4 specific objectives are: 1) **Co-design prevention and collection measures with authorities, industry, and citizen**, 2) Co-develop procurement templates and waste-system improvements with municipalities, 3) **Demonstrate sustainable and safe removal technologies**, 4) Produce operational playbooks and O&M guidelines for public authorities

Description of Work: T4.1 Toolbox architecture & content design - *Lead: CoLab House; Participants: TransfoLAB BCN, Lichen, IEEL; Months: 1-18* This task will define the objectives and structure for the development of guides, tutorials, and training modules. It will identify target user groups and specify functional and non-functional requirements, ensuring alignment with the needs of municipalities, citizens, organisations, and social enterprises. The task will also focus on content selection, categorisation, and the design of a clear information architecture to support usability and relevance. This structured framework will guide the development of accessible and impactful materials in subsequent tasks. D4.1

T4.2 Compilation of guides, playbooks, and methodologies - *Lead: UZAF; Participants: Decode Fab Lab, CoLab House, IDAEA, IEEL, UNIDU; Month: 6-18* This task focuses on the compilation and development of a comprehensive suite of guides, playbooks, and methodologies addressing monitoring, removal, prevention, and management of plastic developed in WP2. The outputs will include practical guides, tutorials, and modular training courses tailored to the specific needs and responsibilities of diverse stakeholder groups within the local ecosystem—namely municipalities, citizens, organisations, social enterprises, and other relevant actors. Each module will be adapted to the role and capacity of the target audience, ensuring relevance and applicability across contexts. A user-centric digital dashboard will be developed by ICCS to facilitate intuitive access to the materials, enabling users to easily navigate, adopt, and apply the content in their respective domains. D4.1

T4.3 Replication & scalability framework (NEB & Green Deal alignment) - *Lead: UZAF; Participants: Comonspace, KYVELI MAE, DUNEA, FLZG; Months: 36-48* This task will produce a comprehensive set of guides, playbooks, and methodologies tailored to different stakeholders. The content will draw on prototypes (WP2), monitoring protocols (WP5), prevention actions (WP6), and education (WP3). Each module will be adapted to specific user groups (e.g. municipalities – procurement playbooks; schools – education toolkits; SMEs/FabLabs – circular entrepreneurship guides). IDAEA will ensure scientific alignment with MSFD TG ML protocols, while Decode and UNIDU contribute to making materials accessible for local training and education. The outputs will be embedded into a user-friendly digital interface developed in WP5, ensuring that all guidance is practical, standardised, and easy to adopt. *Deliverable: D4.1*

Task 4.4: Integration of digital tools and decision dashboards *Lead: ICCS, Participants: TechnAI, ZTKDU; Months: 18–42* The main goal of this task is to combine the digital tools and dashboards made in WP5 with the Toolbox environment made in WP4. The goal is to integrate these digital components into the guides, playbooks and methodologies so that it becomes a seamless end-user experience. This will involve configuring the Toolbox so that decision dashboards can be accessed as functional modules, permitting users to engage live monitoring outputs and forecasts within much of the same framework enclosing those making methodological resources. The Toolbox integrated with basin-level data sharing and comparability will be addressed, including linking to relevant European data infrastructures such as the Digital Twin of the Ocean and EMODnet. D4.2

T4.5 Toolbox validation with stakeholders *Lead: Commonsense, Participants: CoLab House, Lichen, TransfoLAB BCN, ICCS, FLZG, IEET; Months: 30–48* During this task we will evaluate the Toolbox from all aspects, including relevance, accessibility, language and clarity, and whether it is a fit for the target audience. Surveys will be conducted during the activities in WP6 and separate evaluation sessions with stakeholders. During Toolbox validation, feedback on the usability of these dashboards and its fitness for purpose will be collected from municipalities and stakeholders to test post-integration. This provides verification that the digital assets are accessible and operational, and can be used by local/regional mandataries as well citizens with no technical experience. At the end of the task, we will have a fully integrated toolbox package that embeds methodological guidance combined with digital decision-support to be adopted and replicated in pilot regions and Associated Regions. D4.2

Work package number	5 - Lead Partner: ICCS
Work package title	Digital Tools and Platforms for monitoring and open data sharing

WP5 specific objectives are: 1) To design and deploy digital tools and dashboards enabling real-time monitoring, forecasting, and decision support for municipalities and authorities. 2) To develop and maintain an open digital platform hosting the Toolbox, datasets, and methodologies, ensuring accessibility, interoperability, and scalability. 3) To implement harmonised monitoring protocols at pilot sites, aligned with MSFD TG Marine Litter guidelines and EMODnet/DTO requirements. 4) To ensure FAIR data management and open access, linking with WP8 (Data Management Plan) and EU digital infrastructures (DTO, Copernicus, EOSC). 5) To conduct baseline and endline monitoring in pilots (Spain, Greece, Croatia), quantifying plastic flux reductions and validating ecological safety of removal and prevention measures.

Description of Work: Task 5.1: Development of digital tools and dashboards *Lead: ICCS, Participants: TechnAI, ZTKDU, pilot municipalities; Months: 6–36* This task will design and implement the digital tools and data interfaces needed for real-time monitoring of plastic flows, forecasting hotspots, and supporting municipal decision-making. These tools will include modules to capture and standardise heterogeneous inputs from low-cost sensors, drone and computer-vision transects, removal devices, and structured citizen-science applications deployed in the pilot sites. Data acquisition and harmonisation components will be developed with open APIs and a modular microservices architecture, ensuring scalability and smooth integration with the wider Toolbox and EU digital infrastructures. On top of these acquisition and processing layers, the task will deliver user-friendly dashboards that translate raw data into actionable knowledge. Machine-learning models will be embedded to generate short-term forecasts of plastic accumulation hotspots, while geospatial analytics will visualise patterns and long-term trends at both municipal and basin scale. The dashboards will offer functionalities such as early-warning alerts for local authorities, automated compliance reports for regulators, and progress tracking against the Mission Ocean and Zero Pollution targets. All tools and dashboards will be developed in close collaboration with municipalities, schools, and citizen organisations through participatory workshops, ensuring usability, inclusiveness, and relevance to local governance contexts. Interfaces will be customisable for different user groups and aligned with EU public-sector digital standards, including accessibility and multilingual support. By combining data acquisition modules, predictive analytics, and participatory design, this task will deliver a suite of digital monitoring and decision-support tools that are both technically advanced and directly usable by public authorities and communities across the Mediterranean.. D5.1, D5.3

Task 5.2 Digital platform and integration with Toolbox *Lead: ICCS, Participants: TechnAI, ZTKDU; Months: 9–42* This task will implement and manage the open digital platform underpinning the RIPPLE Toolbox. The platform will be a unified, publicly available hub that hosts all project outputs - the Toolbox itself, datasets, guides and playbooks - to ensure their interoperability and future accessibility for different end-users. It will be built as a modular, web-based scalable system that can adapt for new datasets and processes during the duration of project life. By working closely with these efforts, the platform will be able

to interoperate easily leveraging EU digital infrastructures, such as EMODnet, Digital Twin Ocean and EOSC, thereby enabling seamless data flows, compliance with FAIR principles, and alignment with ongoing EU investments in digital knowledge systems. So as to optimize ease of use, the system will have open APIs that can incorporate into municipal and regional governance systems, thereby allowing respective authorities insert monitoring results, removal playbooks & prevention measures right at front lines decisions. User management and access layers will provide tailored interfaces based on the role of municipalities, ARs, NGOs or researchers ensuring transparent yet open access to core datasets and methodologies. In addition, the platform will integrate participation mechanisms (e.g. open-source repositories, feedback modules and community-driven extensions) creating entrepreneur and citizen engagement opportunities as well as promoting innovation beyond the project duration. It will also consist of visualisation and reporting tools focused on the general public - designed for non-technical users (policymakers, educators and civil society organisations). In this way, the platform will act as a unique access point for municipalities, Associated Regions, NGOs, and researchers targeted by adopting, replicating, and scaling up the Toolbox solutions across different Mediterranean geographical contexts.. D5.2, D5.4

T5.3 Monitoring protocols and baseline assessment *Lead CSIC, Participants: TransfoLAB BCN, CoLab House, KYVELI MAE, TeamLabs, ZTKDU, DUNEA, UNIDU; Months: 6-24* CSIC will provide robust and advanced analytical capabilities to ensure accurate detection and quantification of plastic additives and microplastics in environmental samples and marine litter from the pilots, with methodologies designed for upscaling to the Mediterranean basin. Samples collection will be executed jointly with task participants. The analysis will be aligned with T5.4, setting up the baseline for the impact evaluation. D5.5

T5.4 Endline monitoring and impact validation *Lead CSIC, Participants: TransfoLAB BCN, CoLab House, KYVELI MAE, TeamLabs, ZTKDU, DUNEA, UNIDU; Months: 23-45* By applying state-of-the-art liquid and gas chromatography coupled to mass spectrometry (LC-MS and GC-MS) techniques, IDAEA will deliver high-resolution and reproducible chemical characterizations of microplastics, phthalates, organophosphates, and other plastic-related contaminants in water, soil, biota, and litter matrices. The robustness of these analytical protocols will guarantee reliable datasets to support risk assessment, enabling the evaluation of pollutant transport, fate, and associated ecological and human health impacts. Moreover, this analytical precision will allow informed assessments of recycling potential of marine litter. CSIC will lead chemical analyses and ensure methodological transferability from the pilots to the associated regions. Leveraging its strong track record in European project coordination, CSIC will further reinforce the consortium through effective integration of technical outputs, knowledge transfer, training, and dissemination activities. D5.5

Work package number	6 - Lead Partner Commonsplace
Work package title	Local Pilot Programs

WP6 specific objectives are: 1) To demonstrate the Toolbox in three diverse Mediterranean pilot contexts (Spain, Greece, Croatia). 2) To validate prevention, removal, and circular transformation measures under real-life conditions with municipalities and local stakeholders. 3) To test interoperability of monitoring protocols, removal technologies, and circular solutions across different socio-economic and ecological settings. 4) To generate site-level evidence on ecological safety, cost-effectiveness, and societal impact of Toolbox interventions. 5) To build capacity in municipalities, schools, and civic groups for long-term adoption and replication of solutions.

Description of Work : Task 6.1:Pilot preparation & site selection *Lead: Teamlabs, Participants: Commonsplace, TransfoLAB BCN, KYVELI MAE, DUNEA, UNIDU; Months: 18–24* This task will establish the foundations for the three RIPPLE pilots by finalising site selection, aligning governance arrangements, and preparing detailed pilot action plans. The pilots represent diverse Mediterranean contexts in **Spain, Greece, and Croatia** to demonstrate the transferability of the Toolbox across ecological and socio-economic settings. Each pilot will begin with local stakeholder workshops, engaging municipalities, schools, NGOs, and fishing communities to refine site-specific priorities and define roles. Baseline assessments of plastic fluxes, litter composition, and hotspot mapping will be carried out in coordination with WP5 (monitoring protocols), following MSFD TG Marine Litter guidelines to ensure EU compliance. Based on this evidence, integrated pilot action plans will be drafted, including site maps, timelines, and the selection of Toolbox components to be tested (monitoring, removal, prevention, circular transformation). Plans will be reviewed by the Advisory Board and Stakeholder Panel (WP8) to ensure alignment with Mission Ocean and EU Zero Pollution objectives. D6.1, D6.2

Task 6.2 Deployment of removal technologies (booms, traps, robotics, smart bins) *Lead: FLZG, Participants: TransfoLAB BCN, Decode Fab Lab, KYVELI MAE, TeamLabs, DUNEA, municipalities; Months: 24–36* This task will oversee the deployment of removal technologies adapted to the specific contexts of each pilot site. Technologies will include floating booms and litter traps in river and delta systems, robotics for targeted removal, and smart bins for waste collection in coastal and port areas. Deployment will be carried out in collaboration with municipal waste services and environmental agencies to ensure integration into existing management systems. Operational protocols will be tested, including installation, maintenance, and safety procedures. Performance will be monitored in collaboration with WP5 using standardised indicators such as cost per kilogram removed, ecological safety, and efficiency benchmarks compared with other EU-funded initiatives. The results will feed into the Toolbox (WP4) and provide operational guidelines and procurement templates for wider replication. *D6.2*

Task 6.3 Implementation of prevention actions (fishing-for-litter, waste-system fixes, citizen campaigns) *Lead: KYVELI MAE, Participants: Commonsense, Lichen, TeamLabs, DUNEA, municipalities, NGOs; Months: 24–42* This task will design and implement prevention actions at each pilot site, co-created with municipalities, NGOs, schools, and citizen groups. Activities will include fishing-for-litter schemes with local fleets to recover plastics at sea, improvements to municipal waste management systems to address leakage points, and citizen campaigns in schools and communities to encourage behavioural change. Prevention measures will be integrated with citizen science protocols developed in WP1 and linked to education modules in WP3 to ensure long-term adoption. Each intervention will be documented in playbooks (WP4), highlighting context-specific challenges, behavioural insights, and enabling factors for replication in ARs. The outcomes will be measured against baseline conditions and reported to demonstrate reductions in plastic leakage and increased citizen engagement. *D6.2*

Task 6.4 Implementation of circular transformation and social entrepreneurship initiatives *Lead: TransfoLAB BCN, Participants: FLZG, CoLab House, Decode Fab Lab, TeamLabs, KYVELI MAE, DUNEA; Months: 18–48* This task will pilot circular transformation and social entrepreneurship initiatives in the three pilot regions. Collected plastics from removal and prevention activities will be channeled into mobile FabLabs and Precious Plastics workshops, enabling local communities to transform waste into new products. Training and co-creation sessions will be organised with civic groups, SMEs, and entrepreneurs to test small-scale business models, stimulate innovation, and generate community products with economic and social value. The initiatives will be monitored to assess technical feasibility, community uptake, and contribution to local entrepreneurship. Lessons learned will be integrated into the Toolbox (WP4) and linked to procurement templates to support future adoption by municipalities. Evidence from these pilots will also inform replication strategies in ARs. *D6.2*

Work package number	7 - Lead Partner FabLab ZG
Work package title	Communication, Dissemination, and Exploitation

WP7 specific objectives are: **WP7 specific objectives are:** 1) To ensure visibility of RIPPLE and its results at European, Mediterranean, and local levels; 2) To disseminate scientific, technical, and policy outputs to relevant target groups; 3) To foster exploitation and uptake of project results by public authorities, SMEs, NGOs, and EU platforms; 4) To support replication and scaling through the ARs scheme; 5) To ensure synergies with Mission Ocean and related EU initiatives for maximum impact.

Description of Work : Task 7.1: Communication strategy & branding *Lead: FabLab ZG, Participants: all; Months: 1–48* This task will establish the RIPPLE Plan for Dissemination, Exploitation and Communication (PDEC, D7.1), with a first version due at M6 and updated annually. It will include analysis of target groups and the definition of core messages, branding, and visual identity (logo, templates, communication toolkit). A public project website will be maintained throughout the project, complemented by a social media strategy (Twitter/X, LinkedIn, YouTube). Audio-visual materials will be produced, including a professional project video and storytelling clips from pilots and ARs. Regular press releases and newsletters will be distributed via partner networks. Localised communication will be supported by pilot sites and ARs, each preparing local communication plans, annexed to the PDEC. This task ensures RIPPLE's alignment with Mission Ocean activities by establishing formal links with relevant initiatives (Plastic Pirates, WATER4ALL, Interreg MED). Joint workshops, knowledge exchange events, and coordinated dissemination campaigns will be organised in collaboration with the Mission Implementation Platform. Shared datasets will be aligned where possible, ensuring interoperability and visibility at EU level. Cooperation will amplify RIPPLE's policy and societal impact and ensure complementarity with other European efforts. *D7.1, D7.2, D7.6*

Task 7.2 Dissemination to science, industry, policy & civil society *Lead: TechnAI, Participants: all partners; Months: 1–48* This task covers dissemination of results to the scientific, policy, and practitioner communities. A scientific publication strategy will guide responsibilities, journal selection, and authorship rules, ensuring all outputs are Open Access. Results will be disseminated via peer-reviewed journals, conferences (EGU, ICES, European Maritime Day, MSFD TG ML meetings, Mission Ocean Assembly), and repositories (Zenodo, EMODnet). Policy dissemination will include ≥ 6 policy briefs and policy uptake workshops with regional/national/EU authorities. Civil society dissemination will adapt outputs into accessible formats for NGOs, schools, and communities. *D7.1, D7.5, D7.6*

Task 7.3 Exploitation strategy (policy briefs, procurement & financing pathways) *Lead: UZAF, Participants: all partners; Months: 1–48* Exploitation activities will align Key Exploitable Results (KERs) with expected impacts. An IP and Exploitation Manual (D7.3) will set rules for ownership, access rights, and IP management, building on the Consortium Agreement. Two phases will be implemented: capacity building and validation of KERs (M1–M18), followed by exploitation and sustainability planning (M18–M48). Activities include an Exploitation Strategy Workshop, annual updates to the Results Ownership List (ROL), and a consolidated Exploitation Roadmap (D7.4). Financing and procurement pathways will be prepared for municipalities and SMEs, linking results to EU and national funding programmes. *D7.3, D7.5, D7.6*

Task 7.4 ARs scheme *Lead: UZAF, Participants: all partners; Months: 9–48* This task designs and implements the ARs scheme, including transparent eligibility criteria, application guidelines, and an open call (M9–M12). At least 3 ARs will be selected and supported with cascade funding, mentoring, technical assistance, and access to the Toolbox. Peer-to-peer exchanges with pilots and joint workshops will support replication. Each Associated Region will co-develop a replication roadmap (D7.5) aligned with local policies and funding instruments. *D7.4, D7.5, D7.6*

Work package number	8 - Lead Partner UZAF
Work package title	Coordination and Governance

WP8 specific objectives are: 1) To coordinate and supervise all project research, innovation, and demonstration activities according to the work plan, ensuring quality, timing, and coherence across WPs. 2) To carry out the overall administrative and financial management, including reporting and liaison with the European Commission. 3) To establish efficient internal communication, reporting, and quality assurance mechanisms, covering ethics, risk management, and compliance with Horizon Europe rules. 4) To monitor risk and develop contingency measures for financial, legal, administrative, and technical aspects. 5) To ensure gender balance, inclusion, and Responsible Research and Innovation (RRI) principles across the consortium and stakeholder engagement processes. 6) To oversee data management according to FAIR principles, including preparation and updates of the Data Management Plan (DMP), ensuring open access and compliance with GDPR. 7) To coordinate the Advisory Board and Stakeholder Panel, ensuring external expertise informs project implementation.

Description of Work : Task 8.1: Overall management and coordination *Lead: UZAF, Participants: all partners, M1–M48.* Overall project management will establish and maintain the governance structure to coordinate, communicate, and monitor project progress, guaranteeing the achievement of objectives by efficient use of resources. The governance structure includes: General Assembly (GA): All partners, chaired by the Coordinator, meeting twice per year in person. The GA validates the overall work plan, approves major changes, and provides a platform for collective decision-making. Remote participation will be enabled where needed ; Steering Committee (SC): Composed of WP leaders and chaired by the Coordinator. The SC meets monthly online and twice annually in person to monitor progress, review deliverables and milestones, manage risks, and ensure inter-WP coordination; Management Team (MT): Includes the Project Coordinator, Project Manager, Data Manager, Innovation Manager, Diversity and Inclusion Advisor, and Climate Neutrality Advisor. The MT meets monthly online to provide operational oversight, prepare GA/SC meetings, and ensure quality assurance ; Advisory Board (AB): At least six external experts from science, industry, NGOs, and policy fields will be engaged by Month 6. The AB meets annually and provides independent advice, ensuring that project outcomes are robust, demand-driven, and aligned with Mission Ocean goals. *D8.1, D8.4, D8.5*

Task 8.2: Kick-off and regular project meetings *Lead: UZAF, Participants: all partners, M1–M48.* A consortium-wide kick-off meeting will be held by Month 2 to establish shared understanding of the project structure, tasks, and responsibilities. GA meetings will take place twice annually, rotating among pilot countries (Spain, Greece, Croatia) to enhance partner and stakeholder engagement. SC meetings will be held monthly online, complemented by biannual in-person meetings. Dedicated cross-WP coordination meetings

will be organised quarterly to ensure alignment between scientific/technical work (WP1–WP6) and impact activities (WP7). The GA will meet twice per year (6 times in total), hosted alternately by pilot partners. The SC will meet monthly online (36 times in total) and face-to-face twice per year (6 times in total). The Advisory Board will meet annually (3 times in total). These meetings will ensure cross-partner coordination, risk management, and timely delivery of results. D8.1, D8.4

Task 8.3: Risk and quality management Lead: *UZAF*, Participants: *all partners, M1–M48*.

A Quality Assurance and Risk Management Plan (QARMP) will be delivered in Month 3 and updated annually. It will include procedures for monitoring, internal reporting, peer review of deliverables, risk identification, conflict resolution, and escalation protocols. Initial risks (technical delays, stakeholder disengagement, interoperability barriers, etc.) and mitigation strategies will be reviewed monthly by the SC. An internal project handbook will be developed to guide all partners on quality assurance, communication flows, reporting formats, and templates. D8.2, D8.4

Task 8.4: Data management and ethics Lead: *UZAF*, with *ICCS*, M1–M48.

A FAIR-compliant Data Management Plan (DMP) will be prepared by Month 6, with updates in Months 18 and 30. It will detail data handling, collection, storage, sharing, preservation, and re-use, ensuring alignment with EMODnet, DTO, and EOSC. The Ethics Management Plan (EMP) will also be delivered in Month 6 and updated annually, covering human participants (citizen science), personal data (GDPR compliance), environmental safety (LCA, ecotoxicity protocols), and AI/digital tools. Annual training sessions on ethics and data protection will be held for all partners and pilot stakeholders. All validated datasets will be deposited in trusted repositories (Zenodo, EMODnet, OpenAIRE) with persistent identifiers (DOI). D8.3, D8.4

Task 8.5: Strategic coordination with Mission Ocean and EU initiatives Lead: *UZAF*, Participants: *all partners, M1–M48*. This task ensures active engagement with the Mission Ocean Implementation Platform, sister projects under the same call, and related EU initiatives (e.g., Plastic Pirates, WATER4ALL, Interreg projects). Formal cooperation agreements will be established where relevant, including joint workshops, data-sharing, and dissemination activities. RIPPLE will provide annual contributions to Mission Ocean monitoring reports, demonstrating how project results support the 50% reduction target in plastic pollution. D8.4, D8.5

Table 3.1c: List of Deliverables

No.	Deliverable name	Short description	WP No.	Short name of LP	Type	Dissemination level	Delivery date (in months)
D1.1	Participatory Diagnosis Report	Report on the citizen activities in the three pilot sites (Besòs, Remathia, Neretva), collecting findings and highlighting dreams, fears and expectations.	WP1	Common space	R	PU	M12
D1.2	Stakeholder Collaboration Framework	Framework that defines participation objectives, expectations, principles, decision-making methods, and governance, including gender equality, inclusion of vulnerable groups, and RRI principles.	WP1	Lichen	R	PU	M12
D1.3	Citizen Engagement Strategies and Implementation Report	Engagement strategies, including citizen science protocols, educational resources, behavioural nudges, and awareness campaigns linked to local events and festivals	WP1	Lichen	R	PU	M18
D2.1	Technical Solutions Overview and procurement framework	Report mapping the existing technology for monitoring, collection and circular manipulation of waste. The report will also include an evaluation and	WP2	CoLab House	R	PU	M18

		procurement framework for municipalities.					
D2.2	Technology Assessment and Prototyping Report	Report that will evaluate the outcomes from the co-design, co-creation and prototyping activities within the local contexts in the pilot locations.	WP2	TransfoL AB BCN	R	PU	M18
D3.1	Education and Training modules	Compilation of educational and training material.	WP3	Decode FabLab	DE M	PU	M24
D3.2	Evaluation of Education and Training modules	Report on the evaluation and assessment of the educational and training modules.	WP3	CoLab House	R	PU	M48
D4.1	Toolbox compilation	Compilation of guidebooks, on waste collection, prevention and revalorization, technical equipment documentation, educational programme and entrepreneurial activities, replication and scalability	WP4	TransfoL AB BCN	DE M	PU	M24
D4.2	Toolbox Validation report	Report on the validation activities with the stakeholders	WP4	CoLab House	R	PU	M48
D5.1	Digital tools and dashboard architecture & design specifications	Specifications of system architecture, data acquisition modules (sensors, drones, citizen science, removal devices), data pipelines, APIs, and dashboard design, co-designed with pilot municipalities and stakeholders.	WP5	ICCS	R	PU	M12
D5.2	Final digital tools and dashboards with predictive analytics and reporting functions	Operational digital monitoring tools and dashboards integrating multi-source data, predictive models, geospatial visualisation, alerts, and reporting functions, validated in pilot sites.	WP5	ICCS	DE M	PU	M36
D5.3	Platform architecture and interoperability plan	Technical plan for the development of open platform (to be interoperable with EMODnet, DTO and EOSC and to have local governance system integrated)	WP5	ICCS	R	PU	M24
D5.4	Final Toolbox digital platform	Operational, comprehensive platform for hosting Toolbox modules, datasets and guides/playbooks (with strategic open-source contributors engagement tools & fully working replication features).	WP5	ICCS	DE M	PU	M48
D5.5	Impact assessment report	Report that presents the baseline and endline measurements and analysis of the outcome.	WP5	CSIC	R	PU	M48
D6.1	Pilot site action plans	Site selection and detailed implementation plans for Spain, Greece, and Croatia pilots, including baseline conditions, stakeholder mapping, and local governance arrangements.	WP6	Common space	R	PU	M12

D6.2	Pilot implementation & impact validation report	Consolidated report across all three pilots documenting effectiveness of interventions, monitoring results, ecological safety, and lessons for replication.	WP6	Common space	R	PU	M48
D7.1	PDEC (Plan for Dissemination, Exploitation and Communication)	Planned and implemented communication, dissemination and exploitation activities, including local communication plans (LCPs). Updated annually.	WP7	FabLabZ G	R	PU	M6
D7.2	Communication Toolkit	Visual identity, templates, branding guidelines, and first promotional materials.	WP7	FabLabZ G	R	SEN	M9
D7.3	ARs Open Call Package	Guidelines, eligibility criteria, and communication materials for the ARs open call.	WP7	UZAF	R	PU	M12
D7.4	IP & Exploitation Manual	Guidance on IP challenges, risks, and opportunities; reflects CA rules on access rights and ownership.	WP7	UZAF	R	SEN	M12
D7.5	Exploitation, policy briefs & replication guidance	≥6 policy briefs plus replication roadmaps from ARs, providing recommendations for uptake and scaling.	WP7	UZAF	R	PU	M42
D7.6	Final Dissemination & Exploitation Report	Summary of all DEC activities, achieved KPIs, and sustainability pathways.	WP7	FabLabZ G	R	PU	M48
D8.1	Project management plan & timeline	Defines governance structure, rules, reporting templates and internal communication.	WP1	UZAF	R	SEN	M2
D8.2	Quality Assurance & Risk Plan	Framework for usability, accountability, risk monitoring and quality control of data.	WP1	UZAF	R	SEN	M3
D8.3	Data Management Plan (DMP)	Ensures FAIR principles for data and research outputs; updated in M24 and M42.	WP1	UZAF	R	DMP	M6
D8.4	Final administrative and financial report	Reports between partners and towards funding body.	WP1	UZAF	R	SEN	M12, M24, M36, M48
D8.5	Final administrative & financial report	Final consolidated project report with scientific, financial and impact results.	WP1	UZAF	R	SEN	M48

Table 3.1d: List of milestones

No	Milestone name	WP	Due M	Means of verification
M1	M1 Governance and quality framework established	WP8	M3	D8.1 Project management plan; D8.2 Quality Assurance & Risk Plan & D8.3 DPM approved
M2	Communication and dissemination strategy operational	WP7	M9	D7.1 PDEC adopted & D7.2 Communication Toolkit delivered

M3	Citizen participation baseline completed	WP1	M12	D1.1 Participatory Diagnosis Report & D1.2 Stakeholder Collaboration Framework delivered
M4	Digital tools architecture and pilot baselines validated	WP5	M12	D5.1 Architecture specifications & D6.1 Pilot site action plans completed
M5	Associated Regions open call launched	WP7	M12	D7.3 ARs Open Call Package published
M6	Citizen engagement protocols operational	WP1	M18	D1.3 Citizen Engagement Strategies Report submitted
M7	Technical solutions mapped and prototyped	WP2	M18	D2.1 Technical Solutions Overview & D2.2 Prototyping Report delivered
M8	Education modules available	WP3	M24	D3.1 Education and Training Modules published
M9	Toolbox first version compiled	WP4	M24	D4.1 Toolbox compilation released
M10	Interoperable platform and monitoring framework ready	WP5	M24	D5.3 Platform architecture & interoperability plan submitted
M11	Dashboards and predictive tools operational	WP5	M36	D5.2 Digital tools & dashboards validated in pilots
M12	Policy briefs and replication guidance prepared	WP7	M42	D7.5 Exploitation, Policy Briefs & Replication Guidance published
M13	Pilot implementation consolidated	WP6	M48	D6.5 Pilot implementation & impact validation report submitted
M14	Final Toolbox and digital tools validated and published	WP5	M48	D3.2 Evaluation of Educational and training modules & D4.2 Toolbox validation report are validated and D5.4 Final Toolbox digital platform is fully operational and validated
M15	Project finalised and impact documented	WP8	M48	D7.6 Final Dissemination & Exploitation Report & D8.5 Final project report submitted

Table 3.1e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	Work package(s) involved	Proposed risk-mitigation measures
Management & coordination		
Lack of coordination among partners (Low / High)	WP8, all	Governance structure (GA, SC, MT) defined; monthly SC meetings; RACI matrix; early warning indicators; reallocation of roles if needed.
Budget constraints or cost inflation (Medium / High)	WP8, all	Detailed budgeting; supplier diversification; budget reallocation under HE rules; hybrid meetings to reduce travel costs.
Staff turnover or insufficient capacity (Medium / Medium)	WP8, all	Knowledge captured in shared workspace; shadowing; backfilling from partner networks; SC reassigns subtasks if necessary.
Data collection & interoperability		
Insufficient baseline data or site access delays (Medium / High)	WP5, WP6	Early site agreements with municipalities; seasonal planning buffers; proxy/open data use; extend baseline period if needed.
Data quality or interoperability shortfalls (Medium / High)	WP1, WP5, WP6	SOPs aligned with TG ML; QC/validation scripts; harmonised metadata; DTO/EMODnet schema validation before publication.
GDPR/ethics non-compliance (Low / High)	WP1, WP3, WP5, WP8	Ethics & Data Plans at M6; consent/assent forms; anonymisation; annual training; controlled access to raw data.

Pilot deployment & technical performance		
Permitting or authorisation delays for devices (Medium / High)	WP6	Early permitting checklists; municipal focal points; low-risk devices first; contingency sites identified.
Underperformance or ecological side-effects of removal tech (Medium / High)	WP6, WP4, WP5	Bench testing; no-harm protocols; ecotoxicity checks; swap/retire tech; document lessons in playbooks.
Supply-chain delays or equipment shortages (Medium / Medium)	WP6, WP5	Dual suppliers; buffer stock; early procurement; local fabrication where possible.
Extreme weather damaging installations (Medium / Medium)	WP6	Seasonal deployment; resilient anchoring; insurance; redeployment plan with spare units.
Stakeholder engagement & participation		
Low citizen/school participation (Medium / Medium)	WP1, WP3, WP6, WP7	Co-design with schools/NGOs; micro-grants and recognition; targeted campaigns; adapt activities from feedback.
Weak collaboration among agencies (Medium / High)	WP6, WP4	Local coordination groups; MoUs clarifying roles; facilitator support; escalation to municipal leadership.
Fishing-for-litter uptake lower than expected (Medium / Medium)	WP6, WP1	Incentives (vouchers, recognition events); simple logistics; expand to additional fleets if uptake is low.
Limited policy buy-in for replication (Medium / High)	WP7, WP4	Early engagement; briefs tied to local priorities; peer exchanges; demonstrate cost–benefit quick wins.
FSTP & ARs		
Low response to open call (Medium / Medium)	WP7, WP8	Wide dissemination (Mission Ocean, EMODnet, NCPs); dedicated webinar/Q&A; targeted outreach; extend deadline if needed.
Geographical imbalance of applicants (Medium / Medium)	WP7, WP8	Balance included in selection criteria; targeted outreach in under-represented sub-basins.
Conflict of interest in evaluations (Low / High)	WP7, WP8	Independent experts screened; signed impartiality declarations; ESRs and redress procedure in place.
Sub-grantee under-delivery or non-compliance (Medium / High)	WP7, WP8	Sub-grant agreements with milestones; staged payments; technical mentoring; corrective actions or reallocation to reserve list.
Exploitation & IP		
Low adoption of Toolbox results (Medium / Medium)	WP4, WP7	User-driven design; procurement templates; total cost of ownership benchmarks; demos at EU/multiplier events.
IP or ownership disputes (Low / High)	WP7, WP8	IP Manual; Results Ownership List; CA clauses on background/foreground; joint ownership agreements as needed.
External context		
Pandemic or travel restrictions (Low / Medium)	All	Hybrid meeting model; remote training; local facilitators; reschedule fieldwork windows if needed.
Regulatory changes affecting deployments (Low / Medium)	WP6, WP7	Policy watch via partners; adjust deployments; alternative sites/solutions.
Geopolitical or macroeconomic shocks (Low / Medium)	WP8	Scenario planning in QARMP; diversify suppliers; build contingencies into schedules.

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Table 3.1f: Summary of staff effort

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	Total PM Participant
1/UZAF	5	5	8	15	4	4	16	21	78
2/CoLab	6	12	16	9	7	14	6	2	72
3/COMMONSPACE	8	10	5	5	5	12	5	2	52

4/CSIC	4	0	6	8	23	4	2	2	49
5/KYVELI	11	7	7	8	10	26	12	4	85
6/ICCS	0	0	10	6	25	3	2	2	48
7/DECODE	3	6	8	6	2	12	4	2	43
8/TRANSFOLAB BCN	3	14	12	9	2	17	8	4	69
9/TechnAI	3	0	2	6	16	2	2	2	33
10/FabLabZg	5	11	10	8	0	12	19	4	69
11/Lichen	22	2	0	9	0	4	4	2	43
12/DUNEA	8	7	6	7	8	24	8	4	72
13/IEEI	3	10	4	9	0	0	12	2	40
14/ZTKDU	4	6	0	8	12	2	4	2	38
15/TEAMLABS	3	4	6	6	5	23	6	4	57
16/ UNIDU	2	0	2	2	2	3	3	2	16
Total Person Months	90	94	102	121	121	162	113	61	864

Table 3.1g: ‘Subcontracting costs’ items

10/FLZG		
	Cost (€)	Description of tasks and justification
Subcontracting	15,000	External quality testing of the methodological framework, design processes, and social innovation outputs to guarantee environmental safety, durability, and compliance with EU standards.

Table 3.1h: ‘Purchase costs’ items (travel and subsistence, equipment and other goods, works and services)

1/UZAF		
	Cost (€)	Justification
Travel and subsistence	12,000	Consortium meetings and pilot workshops, including stakeholder events in pilot
Equipment	25,000	Purchase equipment, renting equipment, depreciation cost for
Other goods, works and services	10,000	Workshop materials, venue and logistics for local co-design sessions, dissemination and translation costs, facilitation support.
Total	47,000	

3/COMMONSPACE		
	Cost (€)	Justification
Travel and subsistence	10,000	Consortium meetings and pilot workshops, including stakeholder events in pilots
Equipment	10,000	Small IT and audio-visual devices, plus basic citizen science/monitoring kits to support engagement activities
Other goods, works and services	10,000	Workshop materials, venue and logistics for local co-design sessions, dissemination and translation costs, facilitation support
Total	30,000	

4/CSIC		
	Cost (€)	Justification
Travel and subsistence	16,000	3 consortium meeting for 2 people , 2 trips for conferences for 2 people, sampling campaigns in the different case studies
Equipment	9,000	Equipment for plastic shredder

Other goods, works and services	27,000	Consumables for chemical analysis (solvents, analytical standards, glass materials, columns) and samples extraction
Total	52,000	

5/KYVELI

	Cost (€)	Justification
Travel and subsistence	10,000	Consortium meetings and pilot workshops, including stakeholder events in pilot
Equipment	40,000	Equipment for monitoring (sensors, drones, booms, traps)
Other goods, works and services	12,000	Workshop materials, venue and logistics for local co-design sessions, dissemination and translation costs, facilitation support.
Total	62,000	

10/FLZG

	Cost (€)	Justification
Travel and subsistence	12,000	Consortium meetings and pilot workshops, including stakeholder events in pilot
Equipment	25,000	Purchase equipment, renting equipment, depreciation cost for prototyping & pilot testing of open-source tools
Other goods, works and services	10,000	Workshop materials, venue and logistics for local co-design sessions, dissemination and translation costs, facilitation support.
Total	47,000	

12/DUNEA

	Cost (€)	Justification
Travel and subsistence	18,000	Consortium meetings and pilot workshops, including stakeholder events in pilot
Equipment	30,000	Equipment for monitoring (sensors, drones, booms, traps)
Other goods, works and services	14,000	Workshop materials, venue and logistics for local co-design sessions, dissemination and translation costs, facilitation support.
Total	62,000	

14/ZTK

	Cost (€)	Justification
Travel and subsistence	5,600	Consortium meetings and pilot workshops, including stakeholder events in pilot
Equipment	12,000	Low-tech tools and parts to build them, small tools and consumables for co-creation activities, logistics, dissemination activities, support to pilot, consortium meetings
Total	17,600	

15/TEAMSLAB

	Cost (€)	Justification
Travel and subsistence	16,000	Participation in 7 consortium meetings, local travel for 3 co-design workshops with exchanges with Regions
Equipment	30,000	Limited purchase/depreciation of pilot tools and machines
Other goods, works and services	20,000	Co-organisation of consortium meetings in Barcelona; 3 co-design /stakeholder workshop. Dissemination outreach, translation
Total	66,000	

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